

Diagonal Euler transforms (OEIS A252782 and A270917)

The n^2 -th root tends to $3^{1/3}$, not $e^{1/e}$: exact phases,
residue-aware inverses, the critical crossover and the growing crossover

Two research reports of one external research session
(bundle Reports 147 and 148), merged into one ProveIt report

Both Parts: 3 October 2026 merged: 5 October 2026

Abstract

Let $A_\varepsilon(n) = [x^n] \prod_{j \geq 1} (1 - \varepsilon x^j)^{-\varepsilon j^n}$: for $\varepsilon = +1$ the n th term of the Euler transform of the n th powers, OEIS A252782, and for $\varepsilon = -1$ its distinct-colour analogue, A270917. Both OEIS entries state the conjecture $A_\varepsilon(n)^{1/n^2} \rightarrow \exp(e^{-1}) = 1.44466786\dots$ (Václav Kotěšovec, March 25, 2016). This report prints two manuscripts as Parts I and II. Part I proves that the limit is $3^{1/3} = 1.44224957\dots$, which refutes the conjecture: $\log x/x$ is maximal at $x = e$ over the reals but at $j = 3$ over the integers. The write adds the bound $A_\varepsilon(n) \leq 2^{n-1} 3^{n^2/3}$ with a complete elementary proof, from which $A_\varepsilon(n)^{1/n^2} < e^{1/e}$ for every $n \geq 414$. Part I then gives an exact expansion over logarithmic atoms, the three residue-dependent leading amplitudes with all exponentially small sectors and a uniform tail bound, the first unrestricted-minus-distinct difference, power-log inverses to every order, convergent inverses of finite sector models, asymptotic residue-aware integer threshold brackets, strict increase at every $n \geq 1$, and a uniform crossover for a separately varying exponent τ with bounded $z = m^2(8/9)^\tau$ ($n = 3m + r$), including a correction of non-integer order $m^{-\delta}$, $\delta = 2.0959\dots$, from parts of size five. Part II extends the crossover to growing $z \leq Sm^{3/4}$, where the coefficient acquires the attenuation $\exp(-8s^{4/3}/9)$, $s = z/m^{3/4}$, with two quarter-power corrections and a term of order $m^{-0.6849\dots}$, and inverts the attenuation of a smooth model on compact windows; it answers in part a question Part I left open. Effective constants and onsets, all-orders expansions in the crossover regimes, the range $s \rightarrow \infty$ and integer-exponent thresholds remain open. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The two Parts are two manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 47fc7a069 (batch 106). They study one family of coefficients,

$$A_\varepsilon(n, \tau) = [x^n] \prod_{j \geq 1} (1 - \varepsilon x^j)^{-\varepsilon j^\tau}, \quad \varepsilon \in \{+1, -1\},$$

on the diagonal $\tau = n$ (OEIS A252782 for $\varepsilon = +1$, A270917 for $\varepsilon = -1$) and off it, with $n = 3m + r$, $r \in \{0, 1, 2\}$:

Part	Bundle report	Delivered title	Labels
I	147 (base)	Diagonal Euler transforms: Exact phases critical crossover and residue aware inverses	det:dia:
II	148	Growing Euler crossover and attenuation inverse: Uniform moderate defect clouds for diagonal Euler families	det:grow:

Arrangement. The base of the merge is Report 147; its `Report147.tex` was staged as this `article.tex`, and it is printed first, as Part I. Report 148 extends Part I’s critical crossover (Section 10) from bounded $z = m^2(8/9)^\tau$ to $0 < z \leq Sm^{3/4}$, and so answers in part Part I’s further question 3 (Section 13). It restates the definitions it needs and proves its own normalization and estimates (its Section 25 says so); it takes from Part I the setting, the leading amplitudes and the pure-defect functions Ψ_r . It is printed second, as Part II. Dependency order is also the order of writing.

Embedded copy. Report 148’s archive carries Report 147’s source and PDF under `foundation/`; both files are byte-identical to Report 147’s standalone delivery (SHA-256 at placement, `cmp` again at the write). They are printed once, as Part I, and are not shipped.

Numbering. Sections are numbered continuously, and statements and equations within sections, so a statement or equation number moves with its section only. Part I keeps Report 147’s Sections 1–13 and all its numbers. Report 148’s Section k is Section $k + 13$ here (Sections 14–26), and its statement or equation $k.j$ is $(k + 13).j$: its Theorem 2.1 and Corollary 2.2 are Theorem 15.1 and Corollary 15.2, its Propositions 5.1 and 6.1 are Propositions 18.1 and 19.1, and its Theorems 8.1 and 9.1 are Theorems 21.1 and 22.1. Subsection 2.1 and Section 27 were added in the write; no delivered number moved.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date (5 October 2026). They give cross-references between the Parts, mark the question that Part II answers, mark restated material, and record what the write checked. No delivered statement was changed and no delivered symbol was renamed; the one printed decimal found wrong (a digit of β in Part II) is corrected in a dated note beside it. Each Part ends with its own section of open questions (Sections 13 and 26); Section 27 gathers them for the whole report under Vladimir’s standing rule of 4 October 2026, with their sources, sketches and what is missing.

What is proved, and where

Here $n = 3m + r$; $z = m^2(8/9)^\tau$ (Part II writes t for τ); $s = z/m^{3/4}$; $\rho = 5/\sqrt{32}$ and $\delta = 2 \log(\sqrt{32}/5) / \log(9/8) = 2.0958882 \dots$

Statement	Where
$\log A_\varepsilon(n) = \frac{\log 3}{3}n^2 + O(n \log n)$, so $A_\varepsilon(n)^{1/n^2} \rightarrow 3^{1/3}$: the OEIS conjecture $\rightarrow e^{1/e}$ is false	Part I, Theorem 2.1
$3^{mn}/m^m \leq A_\varepsilon(n) \leq 2^{n-1}3^{n^2/3}$ for $n \geq 3$; $A_\varepsilon(n)^{1/n^2} < e^{1/e}$ for every $n \geq 414$ (and, by exact computation, for every $n \geq 8$)	[write] Proposition 2.2, Corollary 2.3, Remark 2.4
The integer-break maximum $M(n)$ (classical)	Part I, (2.2)
Exact expansion over logarithmic atoms; atom bound $\mu^6 \leq 9^r(8/9)^D$; uniform degree and spectral tails, explicit onset $n \geq 192$	Part I, Theorem 3.1, Lemma 4.1, Theorem 4.2
$A_\varepsilon(n) \sim L_r(n)$ with the three residue amplitudes; every common relative phase $\geq 7/10$ (exhaustive rational certificate)	Part I, Theorem 5.1, Table 1
First unrestricted-minus-distinct difference, phases $4/9, 1/2, 1/2$	Part I, Theorem 6.1, Table 2
Leading inverse with every power-log order; convergent inverse of each finite sector model; compatibility under deeper truncation	Part I, Proposition 7.1, Theorems 7.2 and 8.1, Proposition 8.2
Residue-aware integer threshold brackets (asymptotic, no effective onset); strict increase for every $n \geq 1$	Part I, Theorem 9.1, Proposition 9.2
Uniform crossover for bounded z , entire Ψ_r , two algebraic corrections and the term $\rho^\tau \Phi_r(z)$ of order $m^{-\delta}$	Part I, Theorem 10.1
Growing crossover for $0 < z \leq Sm^{3/4}$: attenuation $e^{-8s^{4/3}/9}$, two quarter-power corrections, a term of order $m^{-\beta}$, $\beta = \frac{5}{8}(\delta - 1)$, remainder $O_S(m^{-3/4})$	Part II, Theorem 15.1, Corollary 15.2
Pure-cloud depletion; the whole bad-atom sum; the size-five term and the zero endpoint	Part II, Proposition 18.1, Lemma 18.2, Proposition 19.1, Section 20
Attenuation inverse of the smooth model on a compact window; exact real-parameter crossings exist and are localized (uniqueness not claimed); a second route to the pure coefficients	Part II, Theorems 21.1 and 22.1, Lemma 23.1
<i>Open</i>	
Effective onsets and certified integer thresholds; all orders in both crossover regimes; the range $s \rightarrow \infty$ and the bridge to fixed exponents; exact monotonicity and integer-exponent sampling; other integer weights; convergence of the inverse series; a full-text literature comparison. See Section 27.	

The finite computations of the two companions (exact integers and rationals) check coefficient rows, identities, enumerations and coefficient algebra; no asymptotic statement rests on them. The inverses are discussed below, after the OEIS statements.

The OEIS conjecture, and its refutation

The two entries as read by the write on 5 October 2026 (A252782 revision #20 of March 22, 2017; A270917 revision #15 of October 16, 2017):

- **A252782** (Alois P. Heinz, Dec 21, 2014), with name
`a(n) = n-th term of Euler transform of n-th powers`
and the formula $a(n) = [x^n] \prod_{j \geq 1} 1/(1 - x^j)^{j^n}$, so $a(n) = A_+(n)$. Its formula section reads, verbatim:

Conjecture: $\lim_{n \rightarrow \infty} a(n)^{(1/n^2)} = \exp(\exp(-1)) = 1.444667861\dots$
 - `_Vaclav Kotesovec_`, Mar 25 2016

- **A270917** (Vaclav Kotesovec, Mar 25, 2016), with name
 Coefficient of x^n in $\text{Product}_{\{k \geq 1\}} (1 + x^k)^{(k^n)}$
 so $a(n) = A_-(n)$. Its formula section reads, verbatim and unsigned, in the entry Kotěšovec
 created on that date:
 Conjecture: $\lim_{n \rightarrow \infty} a(n)^{(1/n^2)} = \exp(\exp(-1)) = 1.444667861\dots$

What is right and what is wrong. The definitions and the listed terms are right: the write recomputed $A_{\pm}(n)$ exactly for $n \leq 109$ (Euler recurrence (11.1)), and the values agree with both entries' b-files, which run to $n = 80$. What is wrong is the conjectured limit. Theorem 2.1 proves $A_{\varepsilon}(n)^{1/n^2} \rightarrow 3^{1/3} = 1.4422495703\dots$, while $e^{1/e} = 1.4446678610\dots$. The write's Subsection 2.1 prints the proof with every constant explicit: from $j \leq 3^{j/3}$ for every integer $j \geq 1$ each of the at most 2^{n-1} profiles has weight at most $3^{n^2/3}$, so $A_{\varepsilon}(n) \leq 2^{n-1} 3^{n^2/3}$; the profile of m parts of size 3 gives $A_{\varepsilon}(n) \geq 3^{mn}/m^m$; and since $\log 3/3 < 1/e$ the upper bound alone gives $A_{\varepsilon}(n)^{1/n^2} < e^{1/e}$ for every $n \geq 414$ (Corollary 2.3; the intake record's " $n \geq 415$ " is valid but one more than needed). So the conjecture is *refuted*, not merely sharpened: the n^2 -th root stays below $e^{1/e}$ from $n = 414$ on and converges to a different number. By an exact finite computation (Remark 2.4) it is below $e^{1/e}$ already for every $n \geq 8$ and above it for $2 \leq n \leq 7$ ($3 \leq n \leq 7$ for A270917). The approach to the limit is slow, $\log A_{\varepsilon}(n)/n^2 = 0.35699$ at $n = 90$ against $\log 3/3 = 0.36620$, which is the $O(\log n/n)$ term of the bounds; Part I explains the two constants by the maximum of $\log j/j$ over the integers ($j = 3$) and over the reals ($x = e$), and says that this "is not evidence about how the original conjecture was produced". Nothing was submitted to the OEIS.

The inverses. None of the inverse statements is an instance of the transseries volume's Lambert core `p0:thm:lambert-core` or factorial core `p0:prop:factorial-core` [10], and no Lambert W occurs in either Part. Statement by statement:

- Part I, Theorem 7.2: the phase $F_r(x) = \alpha x^2 - \frac{x}{3} \log x + c_r x + \dots$ is quadratic, outside the volume's exponential-power model `p0:def:model` (exponent at most 1) and not reduced to it by `p0:lem:alpha-reduction`. Its core αx^2 is an admissible core in the sense of `p0:def:core`, and the triangular recurrence (7.8) is the scheme of `p0:thm:core-reversion` with coefficients that are polynomials in $\log y$: an analogue, proved directly in Part I.
- Part I, Theorem 8.1: the series (8.2) is the series of the volume's perturbed-inversion theorem `p0:thm:perturbed-inversion` with core F_r , perturbation H_r and $\varepsilon = 1$. That theorem gives convergence only for $|\varepsilon| < \varepsilon_0$; Part I proves, for large t , that the disk of convergence contains $|\varepsilon| \leq 2$. So an instance of the formula, with the convergence at $\varepsilon = 1$ proved in Part I.
- Part I, Theorem 9.1 and (9.6): the residue-class ceilings and the minimum over residues have the form of `p0:thm:staircase` (4), and the bracket follows the separation idea of its item (2); but that theorem presupposes an admissible interpolation of the sequence itself, and Part I interpolates only the model ("No interpolation of the original sequence is being assumed"). Not an instance; proved directly at the integers.
- Part II, Theorem 21.1: the core $F(s) = -\gamma s^{4/3}$ is inverted by a power, $s_0 = (-\log q/\gamma)^{3/4}$. The coefficients (21.8) are the first coefficients of the multi-parameter form `p0:eq:perturbed-multi` of `p0:thm:perturbed-inversion` with perturbations G, H_r, J and small parameters u, u^2, v (checked by the write symbolically, including the u^2 coefficient); the remainder is Part II's own mean-value argument. An instance of the coefficient formula.

- Part II, Theorem 22.1: brackets in the real parameter t only; no counterpart in the volume's integer apparatus.

Neither manuscript claims novelty for the inversion mechanics: Part I calls Lagrange inversion classical, and Part II calls its methods classical.

Notation across the two Parts

Common to both Parts, with the same meaning: ε , $n = 3m + r$, $z = m^2(8/9)^\tau$, $a = z^{1/3}$, $\rho = 5/\sqrt{32}$, δ , $H_k = [u^k] \exp(u + u^2)$, Ψ_r , $F_q(a) = \sum_{k \equiv q(3)} H_k a^k$, $\omega = e^{2\pi i/3}$, the atom activities $\lambda_{jk} = j^\tau \xi^{jk}$ (Part II: $\lambda_{j,k}$), the depletion index $\ell = (D - r)/3$, the degree D , and the falling factorial $(m)_\ell$. Part II's L_r is Part I's $\mathcal{L}_r$ of (10.3); Part II's $P_\ell(m)$ is Part I's $W_\ell(m)$; Part II's x_0 is Part I's ξ ; Part II's $q_r = (0, 2, 1)$ and $c_r = (1, 3/2, 1)$ are Part I's κ_r and g_r ; Part II's $\rho_{j,k}$ is Part I's R_{jk} ; Part II's $B = a + a^2$ is Part I's B_0 of Section 10. The symbols below change meaning between the Parts or inside one; each Part's text fixes its meaning there, and each Part opens with a short table of its own readings. No symbol was renamed.

Symbol	Meanings
τ, t	Part I: τ the separately varying outer exponent (Section 10); $t = \log X$ the logarithmic threshold of Sections 7–9. Part II: t the outer exponent (Part I's τ); $t_m(s)$ the exponent at given s ; $\tau_{r,m}(q)$ the inverse value of t (Theorem 21.1). <i>Tempting false reading</i> : Part II's t is not Part I's t .
L	Part I: $L_r(n)$ the diagonal amplitudes (5.1); $L_r(x)$ their smooth branch (7.2); $\mathcal{L}_r$ the two-parameter amplitudes (10.3). Part II: $L = \log(9/8)$; L_r the two-parameter amplitudes (14.4), equal to Part I's $\mathcal{L}_r$ (and to Part I's $L_r(n)$ only on the diagonal $t = n$); L^3 norms. <i>False reading</i> : Part II's L is not an amplitude.
α	Part I: $\alpha = \log 3/3$. Part II: $\alpha = 5/(4L)$ (proof of Proposition 19.1, Section 21).
$\beta, \mathcal{B}$	Part I: β a relative phase (Table 1); β the base in (9.1); $\beta_{\max}$; $\mathcal{B}$ the finite set of sector bases (8.1). Part II: $\beta = \frac{5}{8}(\delta - 1) = 0.6849\dots$ an exponent; $\mathcal{B}$ the set of bad atoms. <i>False reading</i> : Part II's β is an exponent of m , not a base.
B	Part I: $B = T^6/9^r$ (5.6); $B = \max(1, M^{1/6})$ and $B_0 = a + a^2$ (Section 10); $B_k(b)$ Bernoulli polynomials. Part II: $B = a + a^2$; B_0 a bound on the shift b (Lemma 18.2).
q	Part I: $q = \sqrt{2}/3^{1/3}$, $q^6 = 8/9$ (4.1); q the index of $F_q(a)$. Part II: q_r, q'_r residue maps; $q \in Q$ a target level in $(0, 1)$ (Section 21); F_q as in Part I.
κ, g, c	Part I: $\kappa = (0, 2, 1)$ and $g = (1, 3/2, 1)$ (10.16); c_r inverse constants (7.4); $c_\varepsilon(b)$ atom coefficients; $C_r(m)$ first corrections (5.5); $g_{r,k}$ Stirling coefficients. Part II: $\kappa = 1.618\dots$ an exponent (19.2); $\kappa_j(K)$ cumulants; $c_r = (1, 3/2, 1)$; c_1, c_2, c_b inverse coefficients; $g_m(x)$ a Gaussian.
R, ρ	Part I: $R_{jk} = j/2^{jk/2}$; $R_j(z)$ residual coefficients (proof of Theorem 7.2). Part II: $\rho_{j,k}$ for Part I's R_{jk} ; R a radius (proof of Proposition 19.1); R_r^{pure} and $R_{\varepsilon,r,m}(s)$ ratios.
W, P	Part I: $W_\ell(m)$ the cutoff-normalized falling factorial; $P_{\varepsilon,r,\mu}(m)$ equal-phase polynomials (4.3); $P_{r,j}$ inverse polynomials. Part II: $P_\ell(m)$ for Part I's $W_\ell(m)$; $W_\ell = 4w/9$ a variance and W a remainder (Sections 18 and 23). <i>False reading</i> : Part II's W_ℓ is not Part I's $W_\ell(m)$.
Q, E	Part I: $Q_r(m)$ the coefficients of Table 1; $Q_r(\theta)$, $E_r(\theta)$ operators (10.5)–(10.6); E a logarithmic error (9.4). Part II: Q a compact subset of $(0, 1)$; $\mathbb{E}$, $\mathbb{E}_r$ (conditional) expectations.
H	H_k as above (Part I also writes H_t). Part I: $H_r = \log S_r$ (Section 8). Part II: $H_r(s)$ a coefficient function (proof of Theorem 21.1).
$M, \mathcal{M}$	Part I: $M(n)$ the integer-break maximum; $M_r(x)$ the finite model (8.1); M the bound on z in Theorem 10.1. Part II: $\mathcal{M}_{r,m}(s)$ the attenuation model (21.3).

Symbol	Meanings
S, s	Part I: $S_r(x)$ the sector sum; $S_T(B)$ a majorant; $s(b)$ the sign exponent (3.2); $s_1(\ell)$. Part II: S the range constant, $s = z/m^{3/4}$, $\mathcal{S}_{\text{bad}}$.
T	Part I: a rational threshold (5.6); $T = T(M)$. Part II: a radius, T_{bad} , T_0 .
Δ, μ, η	Part I: Δ the bracket half-width (9.2); $\mu(b)$ the phase, μ_* ; η a phase cutoff (Theorem 4.2) and $\eta = 2\log(4/3)/\log(9/8)$. Part II: $\Delta = \ell - \mu_\ell$; $\mu = a + 2a^2$, μ_ℓ ; η the second meaning only.
h, d, p	Part I: h_r, d_r, p_r constants (7.1); $p_{r,\beta}(x)$. Part II: $h = \log(5/4)$ and a moment order; $d = 4a^2/3$, $d_0 = \log(5/(2\sqrt{2}))$; $p = \delta/2 + 5/6$, p_* a probability bound.
u, v, x, y	Part I: u the variable of H_k and a complex parameter (proof of Theorem 8.1); $v_r(t)$ the model inverse; $x_r(t)$ the leading inverse; $y = \sqrt{t/\alpha}$. Part II: $u = m^{-1/4}$, $v = m^{-\beta}$; x_0 the activity radius; y_ℓ .
F, G, J, K	Part I: $F_r(x) = \log L_r(x)$ besides $F_q(a)$; K, J orders. Part II: $F(s) = -\gamma s^{4/3}$, $G, J = s^p$ (proof of Theorem 21.1); $J_m = t_m(I)$; $K = N_2 + 2N_4$ a random variable; K_0, K_1 constants. $\gamma = 8/9$ and the exponents $\gamma_\xi, \gamma_T, \gamma_1, \gamma_{c1}, \gamma_{c2}$ occur in Part II only.

Provenance and merge decisions

Two manuscripts of the session bundle of Reports 1–243. Their author lines read “Report 147” and “Report 148”; neither names an author, a tool or an addressee (each sets an empty PDF author field), neither carries “prepared for private review” wording. They arrived in commit 60f54ea06 and were placed by 47fc7a069 (batch 106, cluster 106-EULERD).

- **Part I**, bundle Report 147, archive `Diagonal_Euler_Transforms_Corrected_Growth_Crossover_and_Inverses_Source.zip` (`Report147.tex`, 1250 lines, 21 pages, dated 3 October 2026); the base. It pins ProveIt commit bb1cb91b7 (Section 12), which exists in the history. Contributes the corrected root limit, the exact atom identity, the uniform tails, the residue amplitudes with the exhaustive sector certificates, the first sign-sensitive difference, the inverses and brackets, strict monotonicity, and the bounded- z crossover.
- **Part II**, bundle Report 148, archive `Growing_Euler_Crossover_and_Attenuation_Inverse_Source.zip` (`Report148.tex`, 1061 lines, 17 pages, dated 3 October 2026); no pin. Its `foundation/` holds Report 147’s source and PDF byte for byte. Contributes the growing crossover with its Poisson conditioning, depletion and bad-atom estimates, the zero endpoint, the attenuation inverse, the exact-crossing localization and the second route to the pure coefficients.

Where the merge had to choose:

1. *Order and base.* Report 147 first, as the base and as the foundation Report 148 builds on (see the Guide).
2. *Restatements.* In its own words Part II restates, from Part I: the two-parameter product (10.1) as (14.1) with t for τ ; the formal logarithm (3.1) as (14.2); the amplitudes (10.3) as (14.4); H_k and Ψ_r (10.4) as (14.5)–(14.6); the critical-scale atom formula (10.15)–(10.16) as (16.3)–(16.5); the real form of the roots-of-unity filter (10.13) as (17.1); the bad-ratio bound, fixed majorant and exceptional (1,1) term of Section 10 ((10.17), (10.18), (10.20)) in Section 20 ((20.6), (20.7)); and the Euler recurrence (11.1) as (24.1). They are printed in full, because Part II’s estimates are stated in its own normalization, with notes pointing to Part I.

3. *Front matter and bibliography.* The two title blocks, abstracts and scope paragraphs are printed at the heads of their Parts; one table of contents replaces two. The two bibliographies are merged into one, keeping every detail of every entry; Report 148's entry for Report 147 now points to Part I. The write added the entry for the transseries volume [10].
4. *Write additions.* This front matter, the reading-conventions tables of the Parts, Subsection 2.1 with Proposition 2.2, Corollary 2.3 and Remark 2.4, Section 27, the dated notes, labels for unlabelled sections, and the label prefixes. Everything else is delivered text.

What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier of batch 106, 5 October 2026) both manuscripts were read in full and no step was found false. Checked there: the key step of the root limit by hand; the first twelve terms of both entries; Theorem 5.1, Table 1 and Theorem 6.1 by exact values to $n = 90$ (the remainders after the displayed sectors, divided by the next base to the power n , stay of polynomial size, and the difference ratio to Theorem 6.1's model is 1.013 at $n = 90$); Proposition 9.2 to $n = 90$; Part II's heuristic scale $d = 4a^2/3$ with $e^{-d^2/(2m)} = e^{-8a^4/(9m)}$, and its inequality $\delta < 11/5$. The write recomputed $A_{\pm}(n)$ for $n \leq 109$ and compared them with the b-files and with $e^{n^2/e}$ (Remark 2.4); checked $L_r(3m + r)$ against (5.1), the Stirling data $c_r, a_r, c_{r,0}, g_{r,k}$ of (7.7) against DLMF (5.11.8), and $P_{r,0}$; recomputed H_k for $k \leq 8$ and the displayed coefficients of Ψ_r ; verified Part II's inverse coefficients (21.8), the logarithmic coefficient in the proof of Theorem 21.1 and the quadratic coefficient $128/81 - 32/81 = 32/27$ symbolically; and recomputed the decimals $3^{1/3}, e^{1/e}, 2/\log(9/8), \delta, \beta, \kappa$ and η to 30 digits, finding one wrong digit (Part II's β , dated note at (15.2)). The delivered suites pass on copies (README; two of Report 148's 26 tests need a POSIX host).

What was *not* done: the analytic estimates (the uniform tails, the Rouché and Lagrange–Bürmann argument, the conditioned Poisson moments, the global falling-factorial approximations, the bad-atom sums, the Gaussian second route) were read and their displayed algebra checked, but they are the manuscripts' proofs, not an independent verification. External inputs: the write read DLMF §5.11, (5.11.2) and (5.11.8). It did *not* read Kotěšovec's arXiv paper [5] (Part I's statement that its pp. 21–22 treat fixed exponents is the manuscript's), and it did not read Arratia–Tavaré [8] or Arratia–DeSalvo [9], of which Part II itself inspected only the abstracts.

Relation to neighbouring reports and formal projects

No other report of the repository treats A252782, A270917, their parent arrays A144048 and A284992, or Euler transforms with exponent equal to the index (repository search, 5 October 2026). The report `a290354-iterated-euler-diagonals`, placed in the same batch, studies m -fold *iterated* Euler transforms by parabolic iteration; it shares no definition, lemma, constant or method with this one, only the words “Euler transform”. Part II's conditioning of independent Poisson components on a weighted sum is the device whose general theory (Arratia–Tavaré) is also used in `a033552-catalan-partitions`; the parallel is of method only. The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex` is related to the inverses as stated above. Placement in the collection confers no formal status: no Lean or Rocq file in the repository treats these sequences, and no statement of this report is formalized.

Part I

Diagonal Euler transforms: exact phases, critical crossover and residue aware inverses

Bundle Report 147, archive `Diagonal_Euler_Transforms_Corrected_Growth_Crossover_and_Inverses_Source.zip`, delivered as `Report147.tex` (21 pages), the base of the merge. Delivered title block “Diagonal Euler transforms / Exact phases critical crossover and residue aware inverses / Report 147 / 3 October 2026”. Printed as delivered, with `[write]` notes; section, statement and equation numbers are those of the delivery. Subsection 2.1 is added in the write.

Reading conventions of Part I	Elsewhere in the report
τ the outer exponent of Section 10; t a logarithmic threshold (Sections 7–9)	Part II: t the outer exponent
$L_r(n)$, $L_r(x)$ amplitudes; $\mathcal{L}_r$ two-parameter amplitudes	Part II: L_r for $\mathcal{L}_r$; $L = \log(9/8)$
$\alpha = \log 3/3$; β a base or relative phase; $q = \sqrt{2}/3^{1/3}$	Part II: $\alpha = 5/(4L)$; $\beta = \frac{5}{8}(\delta - 1)$; q a target level
$W_\ell(m)$; ξ ; $\kappa = (0, 2, 1)$, $g = (1, \frac{3}{2}, 1)$; R_{jk}	Part II: $P_\ell(m)$; x_0 ; q_r , c_r ; $\rho_{j,k}$

Abstract of this Part (as delivered). For the diagonal Euler products defining OEIS A252782 and A270917, the n^2 th-root limit is $3^{1/3}$, correcting the value $\exp(1/e)$ conjectured in the inspected entries. The classical integer-product maximization identifies the dominant part size. An exact logarithmic-atom expansion then gives three residue-dependent leading amplitudes, all exponentially small sectors, and a uniform bound on the entire omitted tail. Exact rational finite certificates enumerate the first sectors and the first unrestricted-minus-distinct difference. We construct every power-log coefficient of each leading inverse, prove convergence of the inverse series for every fixed finite sector model, and transport its error to residue-class integer threshold brackets. The latter are asymptotic brackets, not an unconditional finite-input inversion algorithm. A uniform critical crossover for a separately varying exponent resums the dominant sector ladder and identifies a noninteger finite-size correction caused by size-five parts.

Scope. This is a self-contained mathematical report with an exact standard-library computational companion. The integer-break argument, Stirling expansion, and Lagrange inversion are classical tools. The source comparison is bounded to the inspected sources; no global priority claim or external submission is made.

1 Definitions and the diagonal regime

For $n \geq 0$ and $\varepsilon \in \{+1, -1\}$, define

$$A_\varepsilon(n) = [x^n] \prod_{j \geq 1} (1 - \varepsilon x^j)^{-\varepsilon j^n}. \quad (1.1)$$

Thus A_+ is OEIS A252782 and A_- is A270917 [1, 2]. The products are formal power series: only $j \leq n$ can affect the coefficient. The outer exponent n remains *fixed throughout the coefficient extraction*. In particular, $A_\varepsilon(0) = 1$.

For $\varepsilon = +1$, a part of size j has j^n colors and may be repeated. For $\varepsilon = -1$, each colored part may be used at most once; uncolored sizes may still repeat in different colors. A profile (m_j) with $\sum j m_j = n$ has respective weights

$$\prod_j \binom{j^n + m_j - 1}{m_j}, \quad \prod_j \binom{j^n}{m_j}. \quad (1.2)$$

The second product is interpreted as zero if a lower argument exceeds the upper one.

Throughout, write $n = 3m + r$ with $r \in \{0, 1, 2\}$, and put

$$Z_n = \frac{3^{mn}}{m!}, \quad (m)_{\underline{\ell}} = \prod_{i=0}^{\ell-1} (m - i), \quad (m)_{\underline{0}} = 1. \quad (1.3)$$

A residue-class asymptotic always means $n \rightarrow \infty$ through that class. Constants in error terms may depend on the fixed residue and on a fixed spectral cutoff, but never on n .

1.1 What the source comparison establishes

The two OEIS entries inspected on 3 October 2026 display the conjectured limit $\exp(\exp(-1))$. A252782 attributes it to Václav Kotěšovec on 25 March 2016. The retrieved pages were search-crawled six days before inspection; their site footers are not treated as entry-specific revision histories. Our correction concerns those inspected conjectures.

A144048 is the array whose column k is the Euler transform of j^k ; its main diagonal is A252782 [3]. The referenced q-series paper [5], printed pp. 21–22, treats fixed exponents in weighted partition products. Its fixed-parameter formulas do not supply the uniformity in the exponent needed to set $k = n$. The distinct analogue has parent array A284992 [4].

The elementary distinction is that the diagonal weights select the maximum of $\log j/j$ over *integer* j . Its maximizer is 3. The continuous function has its maximum at e . This explains a possible mechanism for the two different constants; it is not evidence about how the original conjecture was produced. A scoped source and overlap statement appears in Section 12.

[write] **5 October 2026.** Both entries were read again on 5 October 2026 and still display the conjecture; their texts, with signatures and revision numbers, are quoted in the front matter (“The OEIS conjecture, and its refutation”). In A270917 the conjecture line is unsigned, in an entry that Kotěšovec created on March 25, 2016. The b-files of both entries ($n \leq 80$) agree with exact recomputation, so the data are right and only the conjectured limit is wrong. Corollary 2.3 shows that the conjecture is refuted, not merely sharpened.

2 The root limit and classical integer products

Theorem 2.1 (Corrected root limit). *For either sign,*

$$\log A_\varepsilon(n) = \frac{\log 3}{3} n^2 + O(n \log n), \quad \lim_{n \rightarrow \infty} A_\varepsilon(n)^{1/n^2} = 3^{1/3}. \quad (2.1)$$

In particular this limit is different from $\exp(1/e)$.

Proof. For integer $c \geq 1$ and $h \geq 0$, a multiset of h colors can be represented by its sorted word, so $\binom{c+h-1}{h} \leq c^h$; the distinct-color weight is no larger. The integer inequality $j \leq 3^{j/3}$ for every $j \geq 1$

gives, profile by profile,

$$\prod_j \binom{j^n + m_j - 1}{m_j} \leq \prod_j j^{nm_j} \leq 3^{n^2/3}.$$

There are at most 2^n partitions of n , since partitions inject into compositions and the latter number 2^{n-1} when $n \geq 1$. Thus $\log A_\varepsilon(n) \leq (\log 3)n^2/3 + n \log 2$.

For a lower bound take m threes, supplemented by a single one if $r = 1$ or a single two if $r = 2$. For all sufficiently large n , the factor for the threes in both models is at least $(3^n - m)^m/m!$. Hence

$$\log A_\varepsilon(n) \geq mn \log 3 - \log(m!) + O(m^2 3^{-n}) = \frac{\log 3}{3} n^2 + O(n \log n).$$

The padding has positive weight at least one. The upper and lower bounds prove the result. Finally $\log x/x$ has its strict real maximum at e , so $\log 3/3 < 1/e$. $\square$

For orientation only, the two constants are approximately 1.4422495703074083823 and 1.4446678610097661337.

[write] 5 October 2026. Both decimals are correct (recomputed to 30 digits). The proof above uses the integer inequality $j \leq 3^{j/3}$ without proof and states the lower bound “for all sufficiently large n ”; Subsection 2.1 proves the inequality, gives both bounds with explicit constants for every $n \geq 3$, and shows that the upper bound alone keeps $A_\varepsilon(n)^{1/n^2}$ below $e^{1/e}$ for every $n \geq 414$.

The refinement of the dominant profile uses the classical integer-break calculation. If $M(n)$ is the largest product of positive integer parts summing to n , then for $n \geq 2$

$$M(3m) = 3^m, \quad M(3m+1) = 4 \cdot 3^{m-1}, \quad M(3m+2) = 2 \cdot 3^m. \quad (2.2)$$

Indeed, ones can be removed by combining them with another part; a part $j \geq 5$ can be replaced by 3 and $j-3$ with larger product; 4 and $2+2$ have the same product; and $2+2+2$ is inferior to $3+3$. These replacements also identify all maximizing profiles. This argument is classical and is not claimed as new. It motivates the leading terms, but the uniform expansion below supplies their rigorous asymptotic proof.

2.1 [write] The refutation with explicit constants

[write] 5 October 2026. This subsection is added in the write. It restates the bounds of the proof of Theorem 2.1 with every constant explicit, proves the integer inequality $j \leq 3^{j/3}$ that the proof uses, and computes the first index from which the upper bound alone excludes the conjectured value $e^{1/e}$. The quoted OEIS texts are in the front matter, “The OEIS conjecture, and its refutation”.

Proposition 2.2 ([write] Explicit two-sided bound). *For either sign and every $n = 3m + r \geq 3$ (so $m \geq 1$),*

$$\frac{3^{mn}}{m^m} \leq A_\varepsilon(n) \leq 2^{n-1} 3^{n^2/3}. \quad (2.3)$$

Consequently

$$\frac{\log 3}{3} - \frac{2 \log 3}{3n} - \frac{\log(n/3)}{3n} \leq \frac{\log A_\varepsilon(n)}{n^2} \leq \frac{\log 3}{3} + \frac{(n-1) \log 2}{n^2}, \quad (2.4)$$

and $A_\varepsilon(n)^{1/n^2} \rightarrow 3^{1/3}$. The upper bound in (2.3) holds for every $n \geq 1$.

Proof. (i) $j \leq 3^{j/3}$ for every integer $j \geq 1$, with equality only at $j = 3$. For $j = 1$, $1 < 3^{1/3}$. For $j = 2$, $2^3 = 8 < 9 = 3^2$, so $2 < 3^{2/3}$. For $j = 3$ there is equality. If $j \geq 3$ and $j \leq 3^{j/3}$, then $j + 1 = j(1 + 1/j) \leq \frac{4}{3}j < 3^{1/3}j \leq 3^{(j+1)/3}$, because $(4/3)^3 = 64/27 < 3$; so the inequality holds for all $j \geq 3$, strictly for $j \geq 4$.

(ii) *Weights.* For integers $c \geq 1$ and $h \geq 0$, a multiset of h colours from c colours is determined by its nondecreasing word in $\{1, \dots, c\}^h$, so $\binom{c+h-1}{h} \leq c^h$; and $\binom{c}{h} \leq c^h/h! \leq c^h$. Hence each of the two weights in (1.2) of a profile (m_j) with $\sum_j j m_j = n$ is a nonnegative integer at most

$$\prod_j j^{nm_j} = \left(\prod_j j^{m_j} \right)^n \leq \left(\prod_j 3^{jm_j/3} \right)^n = 3^{n \cdot n/3},$$

by (i).

(iii) *Upper bound.* $A_\varepsilon(n)$ is the sum of these weights over the profiles, that is over the partitions of n . Listing the parts of a partition in nonincreasing order is an injection into the compositions of n , of which there are 2^{n-1} for $n \geq 1$. So $A_\varepsilon(n) \leq 2^{n-1} 3^{n^2/3}$.

(iv) *Lower bound.* Take the profile with m parts of size 3, together with one part of size 1 if $r = 1$ or one part of size 2 if $r = 2$. All weights are nonnegative, so $A_\varepsilon(n)$ is at least the weight of this one profile. The padding factor is $\binom{1}{1} = 1$ for a part of size one (one colour) and $\binom{2^n}{1} = 2^n \geq 1$ for a part of size two, in both models. The factor of the threes is $\binom{3^n+m-1}{m} \geq \binom{3^n}{m}$ for $\varepsilon = +1$ and $\binom{3^n}{m}$ for $\varepsilon = -1$. For $1 \leq m \leq N$,

$$\binom{N}{m} = \prod_{i=0}^{m-1} \frac{N-i}{m-i} \geq \left(\frac{N}{m} \right)^m,$$

since $(N-i)/(m-i) \geq N/m$ when $N \geq m > i$. With $N = 3^n \geq m$ this gives $A_\varepsilon(n) \geq 3^{nm}/m^m$.

(v) *Logarithms.* The upper bound gives the right side of (2.4). For the left side, $nm = n(n-r)/3 \geq n(n-2)/3$, and $m \log m \leq \frac{n}{3} \log \frac{n}{3}$ because $1 \leq m \leq n/3$ and $x \log x$ is increasing for $x \geq 1$. Dividing $\log A_\varepsilon(n) \geq nm \log 3 - m \log m$ by n^2 gives the left side. Both sides of (2.4) tend to $\log 3/3$. $\square$

Corollary 2.3 ([write] The conjecture is refuted). *For either sign and every $n \geq 414$,*

$$A_\varepsilon(n)^{1/n^2} \leq 2^{(n-1)/n^2} 3^{1/3} < 2^{1/n} 3^{1/3} < e^{1/e}. \quad (2.5)$$

In particular $\limsup_n A_\varepsilon(n)^{1/n^2} \leq 3^{1/3} < e^{1/e}$, so the limit $e^{1/e}$ conjectured in A252782 and A270917 is false; this needs only the upper bound of Proposition 2.2.

Proof. The first two inequalities are (2.3) and $(n-1)/n^2 < 1/n$. The derivative of $\log x/x$ is $(1 - \log x)/x^2$, so $\log x/x$ has its strict maximum on $(0, \infty)$ at $x = e$, and $\gamma_0 := 1/e - \log 3/3 > 0$. The last inequality of (2.5) is equivalent to $\log 2 < n\gamma_0$. From $1/e > 0.367879441$ and $\log 3/3 < 0.366204097$ we get $\gamma_0 > 0.001675344$, hence $414\gamma_0 > 0.693592$, while $\log 2 < 0.693148$. So the inequality holds for every $n \geq 414$. $\square$

Here $\gamma_0 = 0.0016753449487\dots$ and $\log 2/\gamma_0 = 413.734\dots$, so $n = 414$ is the least index for which $2^{1/n} 3^{1/3} < e^{1/e}$. The sharper middle factor $2^{(n-1)/n^2}$ in (2.5) already gives the conclusion at $n = 413$ ($412 \log 2/413^2 < \gamma_0$), but not at $n = 412$. The intake record states “every $n \geq 415$ ”, which is true but not the least index this bound gives.

Remark 2.4 ([write] Exact comparison for small n). The write computed $A_\pm(n)$ exactly for $0 \leq n \leq 109$ by the Euler recurrence (11.1). The values agree with the b-files of A252782 and A270917 (both $0 \leq n \leq 80$), and a 60-digit comparison of $\log A_\pm(n)$ with n^2/e gives: $A_+(n)^{1/n^2} > e^{1/e}$

exactly for $2 \leq n \leq 7$ and $A_-(n)^{1/n^2} > e^{1/e}$ exactly for $3 \leq n \leq 7$, and $A_\pm(n)^{1/n^2} < e^{1/e}$ for every other n with $1 \leq n \leq 109$. For $110 \leq n \leq 413$, step (iii) of the proof above gives the sharper $A_\varepsilon(n) \leq p(n)3^{n^2/3}$, with $p(n)$ the number of partitions of n , and the exact values of $p(n)$ satisfy $\log p(n) < \gamma_0 n^2$ throughout that range (the margin is smallest at $n = 110$: $\log p(110) = 20.2243 \dots < 20.2717 \dots = 110^2 \gamma_0$; it fails at $n = 109$). With Corollary 2.3, $A_\varepsilon(n)^{1/n^2} < e^{1/e}$ for both signs and every $n \geq 8$. This is a finite computation made by the write, not a computation of the delivered companions. It also shows how slowly the limit is approached: $\log A_\pm(n)/n^2 = 0.356990$ at $n = 90$ and 0.357763 at $n = 109$ (the two signs agree to these digits), against $\log 3/3 = 0.366204$, in line with the $O(\log n/n)$ terms of (2.4).

3 An exact expansion by logarithmic atoms

The formal logarithm of the product in (1.1) is

$$\sum_{j,k \geq 1} \frac{\varepsilon^{k-1} j^n}{k} x^{jk}. \quad (3.1)$$

Call (j, k) an atom, of degree jk . Isolate the core atom $(3, 1)$. A defect profile $b = (b_{jk})$ is a finitely supported nonnegative integer array on the remaining atoms. Define

$$\begin{aligned} D(b) &= \sum_{(j,k) \neq (3,1)} jkb_{jk}, & s(b) &= \sum_{(j,k) \neq (3,1)} (k-1)b_{jk}, \\ c_\varepsilon(b) &= \frac{\varepsilon^{s(b)}}{\prod_{(j,k) \neq (3,1)} k^{b_{jk}} b_{jk}!}. \end{aligned} \quad (3.2)$$

If $D(b) \equiv r \pmod{3}$, set

$$\ell(b) = \frac{D(b) - r}{3}, \quad \mu(b) = \frac{\prod j^{b_{jk}}}{3^{\ell(b)}}. \quad (3.3)$$

Then $\ell \geq 0$ and the phase μ is a positive rational number.

Theorem 3.1 (Exact atom identity). *For every $n = 3m + r \geq 0$,*

$$\frac{A_\varepsilon(n)}{Z_n} = \sum_{\substack{D(b) \leq n \\ D(b) \equiv r \pmod{3}}} c_\varepsilon(b)(m) \underline{\ell(b)} \mu(b)^n. \quad (3.4)$$

Proof. Exponentiate (3.1) and extract a multiplicity b_{jk} from each off-core exponential. The residual degree is $n - D = 3(m - \ell)$ and the core contributes $3^{n(m-\ell)}/(m-\ell)!$. Dividing by Z_n leaves precisely the summand in (3.4). Only finitely many profiles have $D \leq n$, so formal rearrangement is legitimate. The empty defect profile is included when $r = 0$. $\square$

No rising or falling binomial factor has been approximated. In particular, all repeated-color corrections and every sign for the distinct model are already present in the exact identity.

4 Local finiteness and a uniform remainder

Put

$$q = \frac{\sqrt{2}}{3^{1/3}}, \quad q^6 = \frac{8}{9} < 1. \quad (4.1)$$

Lemma 4.1 (Atom bound). *For every off-core atom (j, k) ,*

$$j^2 \leq 2^{jk}, \quad \mu(b) \leq 3^{r/3} q^{D(b)}, \quad \mu(b)^6 \leq 9^r (8/9)^{D(b)}. \quad (4.2)$$

Equality in the first inequality occurs only at $(2, 1)$ and $(4, 1)$.

Proof. For $k = 1$ check $j = 1, 2, 4$. For $j \geq 4$, induction follows from $((j+1)/j)^2 \leq 25/16 < 2$. The excluded value $j = 3$ satisfies the inequality as soon as $k \geq 2$. Increasing k only weakens it. Multiplication over the profile and use of $3\ell = D - r$ prove the remaining inequalities. $\square$

Consequently only finitely many profiles have phase at least any fixed $\eta > 0$: their degrees are bounded, and there are finitely many atoms of bounded degree. Equal-phase coefficients

$$P_{\varepsilon, r, \mu}(m) = \sum_{\substack{b: \mu(b) = \mu \\ D(b) \equiv r \pmod{3}}} c_{\varepsilon}(b)(m)_{\underline{\ell}(b)} \quad (4.3)$$

are finite rational polynomials. For the negative sign they may cancel; zero polynomials can be discarded. The unsigned profile phases are locally finite away from zero.

Local finiteness alone would not control the growing number of terms in (3.4). The following absolute majorant is the necessary uniform step. Let C_D be the sum of $|c_{\varepsilon}(b)|$ over all degree- D profiles, without a residue restriction. Then

$$\sum_{D \geq 0} C_D x^D = \exp \left(\sum_{(j,k) \neq (3,1)} \frac{x^{jk}}{k} \right).$$

At $x = 1/2$ its exponent is at most

$$\sum_{j \geq 1} \frac{2^{-j}}{1 - 2^{-j}} \leq 2 \sum_{j \geq 1} 2^{-j} = 2,$$

so

$$C_D \leq e^2 2^D. \quad (4.4)$$

On the integer summands of (3.4), $0 \leq (m)_{\underline{\ell}} \leq n^{\underline{\ell}} \leq n^{D/3}$.

Theorem 4.2 (Uniform degree and spectral tails). *For every fixed integer $D_0 \geq 1$ and every integer $n \geq 192$, the absolute contribution in (3.4) of degrees $D \geq D_0$ is at most*

$$2e^2 2^{D_0} n^{D_0/3} \left(3^{r/3} q^{D_0} \right)^n. \quad (4.5)$$

Fix a phase cutoff $\eta > 0$. Let μ_ be the largest unsigned profile phase strictly below η , and choose an integer $D_0 \geq 1$ with $3^{r/3} q^{D_0} < \mu_*$. For $n \geq \max(192, D_0)$,*

$$\left| \frac{A_{\varepsilon}(n)}{Z_n} - \sum_{\mu \geq \eta} P_{\varepsilon, r, \mu}(m) \mu^n \right| \leq e^2 2^{D_0} (D_0 + 2) n^{D_0/3} \mu_*^n. \quad (4.6)$$

The sum over phases is finite. This supplies the full exponential-polynomial expansion to every fixed exponential depth.

Proof. Equations (4.2) and (4.4) bound the degree- D absolute contribution by $e^2 3^{rn/3} [2n^{1/3} q^n]^D$. If $2n^{1/3} q^n \leq 1/2$, its sum over $D \geq D_0$ is (4.5). Extending the actual finite sum to infinity only increases the bound.

Here is a safe explicit onset. The displayed condition is equivalent to $4096n^2(8/9)^n \leq 1$. Since $(8/9)^6 < 1/2$, at $n = 192$ its left side is less than $9/256$. Moreover $(1 + 1/n)^2(8/9) < 1$ for every $n \geq 17$, as is seen already at 17 and then by monotonicity. Thus the condition holds for all $n \geq 192$. This onset is sufficient, not sharp.

There are phases below η , for example profiles made of arbitrarily many $(1,1)$ atoms in the required residue. Local finiteness shows that their supremum is attained and positive: start with any one such phase, then only finitely many can exceed it. Every retained profile has degree $D < D_0$ by (4.2). Its term is therefore eligible for (3.4) once $n \geq D_0$. The omitted terms of degree below D_0 have phase at most μ_* and absolute coefficient mass at most $e^2 \sum_{D < D_0} 2^D$. Bound their factorial polynomials by $n^{D_0/3}$ and add (4.5). The resulting bound is smaller than the convenient, deliberately loose constant in (4.6). $\square$

5 Residue amplitudes and the first common sectors

Define

$$\begin{aligned} L_0(n) &= \frac{3^{mn}}{m!} & (n = 3m), \\ L_1(n) &= \frac{3}{2} \frac{4^n 3^{(m-1)n}}{(m-1)!} & (n = 3m+1), \\ L_2(n) &= \frac{2^n 3^{mn}}{m!} & (n = 3m+2). \end{aligned} \tag{5.1}$$

Only sufficiently large m is intended in these asymptotic models.

Theorem 5.1 (Leading equivalents and first sectors). *For either sign, $A_\varepsilon(n) \sim L_r(n)$. More precisely, for fixed nonnegative exponents K supplied by Theorem 4.2,*

$$\frac{A_\varepsilon(3m)}{L_0(3m)} = 1 + \frac{7}{6}(m)_2(8/9)^n + \frac{331}{720}(m)_4(64/81)^n + O(n^K(20/27)^n), \tag{5.2}$$

$$\frac{A_\varepsilon(3m+1)}{L_1(3m+1)} = 1 + \frac{9}{20}(m-1)_2(8/9)^n + \frac{2}{3}(m-1)(5/6)^n + O(n^K(64/81)^n), \tag{5.3}$$

$$\frac{A_\varepsilon(3m+2)}{L_2(3m+2)} = 1 + \frac{25}{24}(m)_2(8/9)^n + m(5/6)^n + O(n^K(64/81)^n). \tag{5.4}$$

In each equation n is the argument on the left. The first relative correction is asymptotic to $C_r(m)(8/9)^n$, where

$$(C_0, C_1, C_2) = \left(\frac{7}{6}(m)_2, \frac{9}{20}(m-1)_2, \frac{25}{24}(m)_2 \right). \tag{5.5}$$

5.1 An exhaustive rational certificate

Before proving the theorem, we specify a finite enumeration with proved bounds; it excludes hidden intervening phases. For residue r and rational absolute threshold $T > 0$, put

$$B = \frac{T^6}{9^r}, \quad u(j, k) = \frac{j^6}{9^{jk}}. \tag{5.6}$$

For the thresholds used below $B \leq 1$. Let $D_{\max}$ be the largest integer with $(8/9)^{D_{\max}} \geq B$. Lemma 4.1 implies that every qualifying profile has $D \leq D_{\max}$. Also

$$\prod_{j,k} u(j,k)^{b_{jk}} = \frac{\mu(b)^6}{9^r} \geq B. \quad (5.7)$$

All off-core $u(j,k)$ lie strictly below one. Thus each atom in a qualifying profile must itself satisfy $u(j,k) \geq B$.

Enumerate every divisor j of each $d \leq D_{\max}$, set $k = d/j$, exclude $(3,1)$, and retain atoms with $u \geq B$. Recurse through their nonnegative multiplicities in a fixed order. Prune a branch only if its degree exceeds $D_{\max}$ or its decreasing product of u is below B . At the leaves impose $D \equiv r \pmod{3}$ and $\mu \geq T$. Every qualifying profile has a unique path; every pruning condition is a necessary-condition failure. All arithmetic, including stopping tests, is rational or integer. This proves exhaustiveness, not merely numerical plausibility.

For common relative phases at least $7/10$, the leading absolute phases are $(1, 4/3, 2)$, so use $T = (7/10, 14/15, 7/5)$. The certificate gives:

r	$D_{\max}$	Eligible atoms (j, k)	Profiles
0	18	$(2, 1), (4, 1), (5, 1)$	14
1	22	$(1, 1), (2, 1), (4, 1), (5, 1), (6, 1)$	21
2	20	$(1, 1), (2, 1), (4, 1), (5, 1)$	17

No eligible atom has $k \geq 2$, so the two signs agree throughout this entire range. The complete aggregates are in Table 1. Its normalization is deliberately stated precisely: each row contributes $Q_r(m)\beta^n$ to $A_\varepsilon(n)/(Z_n\rho_r^n)$, where $\rho = (1, 4/3, 2)$. To normalize by L_r , divide the residue-one rows by $3m/2$; no additional division is needed in residues zero or two.

Table 1: Every common relative phase at least $7/10$

r	β	$Q_r(m)$
0	1	1
0	$8/9$	$\frac{7}{6}(m)_2$
0	$64/81$	$\frac{331}{720}(m)_4$
0	$20/27$	$\frac{3}{2}(m)_3$
0	$512/729$	$\frac{5357}{72576}(m)_6$
1	1	$\frac{3}{2}m$
1	$8/9$	$\frac{27}{40}(m)_3$
1	$5/6$	$(m)_2$
1	$64/81$	$\frac{1979}{13440}(m)_5$

r	β	$Q_r(m)$
1	3/4	1
1	20/27	$\frac{25}{24}(m)_{\underline{4}}$
1	512/729	$\frac{223117}{13305600}(m)_{\underline{7}}$
2	1	1
2	8/9	$\frac{25}{24}(m)_{\underline{2}}$
2	5/6	m
2	64/81	$\frac{1303}{5040}(m)_{\underline{4}}$
2	20/27	$\frac{7}{6}(m)_{\underline{3}}$
2	512/729	$\frac{19093}{518400}(m)_{\underline{6}}$

Proof of Theorem 5.1. The certificate identifies the maximal phase and coefficient in each residue as $(1, 1)$, $(4/3, 3m/2)$, and $(2, 1)$ in the Z_n normalization. These give (5.1). In residue one, one four and two twos have equal product; their atom coefficients 1 and $1/2$ add to $3/2$. Theorem 4.2, applied just below each successive phase, bounds all later phases uniformly and proves both the equivalents and the displayed error terms.

The coefficients can also be derived transparently. Set $H_t = [z^t] \exp(z + z^2) = \sum_{a+2b=t} 1/(a!b!)$. A profile made of twos and fours with a twos and b fours has degree $2t$ and numerator product 2^t . The full $(8/9)^{hn}$ ladder in A_ε/L_r therefore has coefficients

$$H_{3h}(m)_{\underline{2h}}, \quad \frac{2}{3}H_{2+3h}(m-1)_{\underline{2h}}, \quad H_{1+3h}(m)_{\underline{2h}} \quad (5.8)$$

in residues zero, one, and two respectively. At $h = 0$ all equal one. The values

$$H_3 = \frac{7}{6}, \quad H_4 = \frac{25}{24}, \quad H_5 = \frac{27}{40}, \quad H_6 = \frac{331}{720}, \quad H_7 = \frac{1303}{5040}, \quad H_8 = \frac{1979}{13440}$$

recover the first two ladder levels. One five creates further phases: with degree-four padding it gives $(3/2)(m)_{\underline{3}}(20/27)^n$ in residue zero; with one two it gives $(2/3)(m-1)(5/6)^n$ in residue one; alone it gives $m(5/6)^n$ in residue two. The exhaustive enumeration proves no intervening phase has been missed. $\square$

The spectrum is therefore not a single geometric family. In particular, $5/6 > 64/81$ in residues one and two, even though a polynomial prefactor can reverse their numerical size at moderate n .

6 The first unrestricted and distinct difference

Theorem 6.1 (First sign-sensitive sector). *As $n = 3m + r \rightarrow \infty$,*

$$\frac{A_+(n) - A_-(n)}{L_r(n)} \sim \begin{cases} \frac{5}{2}(m)_{\underline{2}}(4/9)^n, & r = 0, \\ \frac{4}{3}(1/2)^n, & r = 1, \\ (1/2)^n, & r = 2. \end{cases} \quad (6.1)$$

Proof. Subtracting the two exact atom identities retains exactly the profiles with odd $s(b)$, each with coefficient $2/\prod k^{b_{jk}}b_{jk}!$. Every eligible integer summand is nonnegative, so cancellation cannot hide a larger odd-profile phase. Apply the exhaustive rational algorithm to the absolute thresholds $T = (4/9, 2/3, 1)$. It gives $D_{\max} = (41, 39, 37)$, and the union of eligible atom lists is

$$(1, 1), (1, 2), (2, 1), (2, 2), (4, 1), (5, 1), (6, 1), (7, 1).$$

After imposing odd parity, exactly 3, 2, 1 profiles remain in the three residues. They are displayed in Table 2. The last column is the coefficient *after* subtracting signs, before multiplication by $(m)_{\underline{\ell}}\mu^n$.

r	D	Off-core atoms with multiplicity	μ	Coefficient
0	6	(2, 1), (2, 2)	4/9	1
0	6	(1, 2), (4, 1)	4/9	1
0	6	(1, 2), 2(2, 1)	4/9	1/2
1	4	(2, 2)	2/3	1
1	4	(1, 2), (2, 1)	2/3	1
2	2	(1, 2)	1	1

Table 2: All odd profiles at or above the leading difference thresholds

The sums in the Z_n normalization are $(5/2)(m)_{\underline{2}}(4/9)^n$, $2m(2/3)^n$, and 1. Division by (5.1) gives (6.1). All remaining odd phases are strictly smaller; the same unsigned majorant as in Theorem 4.2, restricted to odd profiles, controls their sum. A strict spectral gap dominates all fixed polynomial factors, proving the equivalents. $\square$

This proof uses a complete finite certificate for the maximal odd phases, rather than an unsupported informal list of candidate defects.

7 Canonical leading inverses to every power-log order

Put $\alpha = \log 3/3$. For $r = 0, 1, 2$, define the triples

$$(d_r, p_r, h_r) = (0, 1, 1), (4, 4, 3/2), (2, 2, 1), \quad b_r = 1 - d_r/3, \quad \lambda_r = \alpha d_r - \log p_r. \quad (7.1)$$

The canonical smooth leading branch is

$$L_r(x) = \frac{h_r \exp(\alpha x^2 - \lambda_r x)}{\Gamma(x/3 + b_r)}, \quad F_r(x) = \log L_r(x). \quad (7.2)$$

For sufficiently large x this is positive, and substitution $x = 3m + r$ reproduces (5.1) exactly. No interpolation of the original sequence is being assumed.

Proposition 7.1 (Leading inverse). *For sufficiently large t there is a unique large root $x_r(t)$ of $F_r(x) = t$. Its derivative satisfies $x'_r(t) = 1/F'_r(x_r(t))$, with*

$$F'_r(x) = 2\alpha x - \lambda_r - \frac{1}{3}\psi(x/3 + b_r) \sim 2\alpha x. \quad (7.3)$$

Proof. The digamma asymptotic $\psi(z) = \log z + O(1/z)$ on the positive ray [6] gives eventual strict increase. Also $F_r(x) \rightarrow \infty$. Existence, uniqueness on an eventual ray, and the derivative formula follow. $\square$

Let

$$y = \sqrt{t/\alpha}, \quad c_r = (\log 3 + 1)/3 - \lambda_r, \quad a_r = \frac{1}{2} - d_r/3, \quad c_{r,0} = a_r \log 3 - \frac{1}{2} \log(2\pi) + \log h_r. \quad (7.4)$$

Theorem 7.2 (All algebraic-logarithmic orders). *For every fixed $J \geq 0$, there are uniquely specified polynomials $P_{r,j}$, with $\deg P_{r,j} \leq j + 1$, such that*

$$x_r(t) = y + \sum_{j=0}^J \frac{P_{r,j}(\log y)}{y^j} + O\left(\frac{(\log y)^{J+2}}{y^{J+1}}\right). \quad (7.5)$$

They are determined by the recurrence below; in particular

$$P_{r,0}(z) = \frac{z}{6\alpha} - \frac{c_r}{2\alpha}. \quad (7.6)$$

Proof. Let $B_k(b)$ denote the Bernoulli polynomial and set

$$g_{r,k} = \frac{(-1)^{k+1} 3^k B_{k+1}(b_r)}{k(k+1)}.$$

Shifted Stirling expansion with a remainder after each fixed order [6] gives

$$F_r(x) \sim \alpha x^2 - \frac{x}{3} \log x + c_r x - a_r \log x + c_{r,0} - \sum_{k \geq 1} g_{r,k} x^{-k}. \quad (7.7)$$

After $P_0, \dots, P_{j-1}$ have been chosen, substitute $X_{j-1} = y + \sum_{i < j} P_i(\log y) y^{-i}$ in the formal right side of (7.7). Let $R_j(z)$ be the coefficient of y^{1-j} in $F_r(X_{j-1}) - \alpha y^2$. Define

$$P_j(z) = -R_j(z)/(2\alpha). \quad (7.8)$$

For $j = 0$ use $X_{-1} = y$. This also yields (7.6). A new term $P_j y^{-j}$ changes the highest uncanceled order only through multiplication by $2\alpha y$, so the procedure is triangular and unique. Formal expansions of $\log(1+u)$ and $(1+u)^{-k}$ show inductively that its degree is at most $j+1$.

At the J th stage the residual is $O((\log y)^{J+2} y^{-J})$. Fixed-order Stirling remainders justify these substitutions for real large y . The leading equation places the root in an interval $y + O(\log y)$: evaluate F_r at $y \pm C \log y$ with C sufficiently large. On that interval $F'_r \geq cy > 0$. The mean-value theorem divides the residual by this lower bound and proves (7.5). $\square$

For a readily checkable next coefficient,

$$P_{r,1}(z) = -\frac{\alpha P_{r,0}(z)^2 - \frac{1}{3} P_{r,0}(z)(z+1) + c_r P_{r,0}(z) - a_r z + c_{r,0}}{2\alpha}. \quad (7.9)$$

The residue changes already affect the constant term of $P_{r,0}$ through $\lambda_r/(2\alpha)$. A single residue-blind smooth inverse would lose these shifts. Equation (7.5) is an all-orders asymptotic expansion; convergence of this infinite power-log series is not claimed.

8 Convergent inverses of finite sector models

Fix one residue and any fixed forward cutoff that includes the leading phase. After replacing m by $(x - r)/3$ and dividing by $L_r(x)$, the finite model has the form

$$M_r(x) = L_r(x)S_r(x), \quad S_r(x) = 1 + \sum_{\beta \in \mathcal{B}} p_{r,\beta}(x)\beta^x. \quad (8.1)$$

Here $\mathcal{B}$ is a finite subset of $(0, 1)$ and each $p_{r,\beta}$ is a rational function, regular on an eventual positive ray, whose fixed derivatives have polynomial growth. In residue one division by $3m/2$ can create negative powers. Put $H_r = \log S_r$, choosing the branch near 1. The model is positive and strictly increasing for all sufficiently large real x .

Theorem 8.1 (Actual model inverse and first displacement). *Let $v_r(t)$ be the unique large solution $M_r(v) = e^t$, and write $x = x_r(t)$. For every sufficiently large fixed t the series*

$$v_r(t) = x_r(t) + \sum_{k \geq 1} \frac{(-1)^k}{k!} \left(\frac{d}{dt} \right)^{k-1} \left[x'_r(t) H_r(x_r(t))^k \right] \quad (8.2)$$

converges absolutely. If $\mathcal{B}$ is nonempty and $\beta_{\max} = \max \mathcal{B}$, then for some fixed K ,

$$v_r(t) - x = -\frac{H_r(x)}{F'_r(x)} + O\left(x^K \beta_{\max}^{2x}\right). \quad (8.3)$$

If no nonleading sector is retained, $v_r = x_r$ exactly.

Proof. We suppress the residue. Since $S = 1 + o(1)$, the function H and each fixed derivative are bounded by a polynomial times $\beta_{\max}^x$. Thus $(\log M)' = F' + H' \sim 2\alpha x$, proving eventual monotonicity.

For fixed large t consider the complex disk $|z - x| \leq 1$. The Gamma argument is far from its poles; choose an analytic logarithm of Gamma there. The rational amplitudes are also analytic there, and S is uniformly close to one, so $\log S$ has an analytic branch. Uniformly on the disk $F''(z) = O(1)$ and

$$F(z) - t = F'(x)(z - x) + O(|z - x|^2).$$

Rouché's theorem, comparing with $F'(x)(z - x)$ on the boundary, shows that $F(z) - t$ has one zero inside and that its boundary modulus is at least cx . Uniformly $H(z) = O(x^K \beta_{\max}^x)$. Applying Rouché again, $F(z) + uH(z) - t$ has exactly one root in the disk for all complex $|u| \leq 2$. Its z derivative is nonzero throughout the disk for large t , because it equals $F'(x) + O(1) + uH'(z)$. The local implicit roots therefore join to a single analytic function $z(u)$ on a neighborhood of this closed u disk.

Set $w = F(z) - t$. The local inverse of F gives $w = -uH(x_r(t + w))$. Lagrange–Bürmann inversion applied to $z = x_r(t + w)$ yields the Taylor coefficients in (8.2). Analyticity in a disk of radius greater than one proves absolute convergence at $u = 1$.

Convergence alone does not prove the squared-phase remainder in (8.3). To establish that estimate separately, write $A = F'(z) + uH'(z)$. Implicit differentiation gives

$$z' = -\frac{H(z)}{A}, \quad z'' = -\frac{(F''(z) + uH''(z))(z')^2 + 2H'(z)z'}{A}. \quad (8.4)$$

Uniformly for $|u| \leq 1$, $|A| \geq cx$. The defining equation and the above linear approximation show $|z(u) - x| = O(x^{K-1}\beta_{\max}^x)$. Consequently H, H', H'' at $z(u)$ remain polynomial times $\beta_{\max}^x$. Equation (8.4) now gives $z'' = O(x^{K'}\beta_{\max}^{2x})$ uniformly. Taylor's integral remainder along $0 \leq u \leq 1$, and $z'(0) = -H(x)/F'(x)$, prove (8.3). $\square$

For example, the second term of (8.2) is

$$\frac{H(x)H'(x)}{F'(x)^2} - \frac{F''(x)H(x)^2}{2F'(x)^3}. \quad (8.5)$$

Expanding $\log S$ and collecting products of the finitely many bases in (8.2) produces every inverse exponential sector. Products of bases in $(0, 1)$ are locally finite above any positive floor: a product with k factors is at most $\beta_{\max}^k$, bounding k . Equal product phases must be combined, notably $(8/9)^2 = 64/81$. Each coefficient has an all-orders power-log expansion from Theorem 7.2 and differentiation. Such differentiation is justified by the analytic implicit functions and uniform fixed-order Stirling expansions on small complex neighborhoods of the positive ray, equivalently by Cauchy estimates there.

The first forward correction gives the leading inverse displacement

$$-\frac{C_r((x-r)/3)(8/9)^x}{F'_r(x)}, \quad x = x_r(t). \quad (8.6)$$

Its next error base is $64/81$ in residue zero and $5/6$ in residues one and two, with polynomial factors. One must not replace β^x by β^y without also transporting $x - y = O(\log y)$: that shift changes the amplitude by a power of y and a residue-dependent factor.

8.1 Compatibility under deeper truncation

Proposition 8.2. *Suppose two fixed finite normalized models differ by $O(x^K\beta_*^x)$ near positive large x , with $0 < \beta_* < 1$. Their inverse branches satisfy*

$$v_2(t) - v_1(t) = O\left(x_r(t)^{K-1}\beta_*^{x_r(t)}\right). \quad (8.7)$$

Proof. Both inverse branches equal $x_r(t) + o(1)$. At v_1 the difference $\log M_2 - \log M_1$ has the stated forward order, since both normalized models tend to one. The mean-value theorem divides it by the eventual lower derivative bound $cx_r(t)$. $\square$

Thus every fixed inverse spectral depth is realized by an actual invertible finite smooth model and a proved next-rate error. The models are compatible as the cutoff deepens. This does not assert a canonical convergent infinite smooth interpolation of the discrete sequences.

9 Residue aware integer threshold brackets

Let a_n be either $A_+(n)$ or $A_-(n)$. Fix an eventual residue branch and a finite model for which, for all sufficiently large $n \equiv r$ modulo 3,

$$\left| \frac{a_n}{M_r(n)} - 1 \right| \leq Cn^K\beta^n, \quad C > 0, K \geq 0, 0 < \beta < 1. \quad (9.1)$$

The forward spectral theorem supplies such a bound after division by the positive leading amplitude and enlargement of constants. Define

$$\lceil z \rceil_r = r + 3 \left\lceil \frac{z - r}{3} \right\rceil.$$

For a threshold $X = e^t$, let $v = v_r(t)$ and put

$$\Delta = \frac{4C2^K}{\alpha\beta^4} v^{K-1}\beta^v. \quad (9.2)$$

Define $N_r(X)$ to be the least nonnegative index in residue class r with $a_n \geq X$. The global monotonicity proof below shows that no unknown finite initial maximum is needed for these diagonal sequences.

Theorem 9.1 (Asymptotic integer bracket). *For all sufficiently large X ,*

$$\lceil v - \Delta \rceil_r \leq N_r(X) \leq \lceil v + \Delta \rceil_r. \quad (9.3)$$

When $\Delta < 3/2$, the two bounds are equal or are adjacent points of the residue lattice. After all stated hypotheses have been verified at a particular threshold, equal bounds determine the crossing; otherwise one exact evaluation at the lower candidate resolves it.

Proof. For $|n - v| \leq 4$ and $v \geq 8$, the right side of (9.1) is at most $C2^K\beta^{-4}v^K\beta^v$, which is eventually at most $1/2$. The logarithmic error is therefore at most

$$E = 2C2^K\beta^{-4}v^K\beta^v. \quad (9.4)$$

Eventually $(\log M_r)' \geq \alpha v$ throughout $[v - 4, v + 4]$. At displacement Δ , the model logarithm changes by at least $\alpha v \Delta = 2E$. Hence every lattice point in that interval at or below $v - \Delta$ is strictly below the threshold, and every one at or above $v + \Delta$ is strictly above it. The lattice point just below $\lceil v - \Delta \rceil_r$ and the point $\lceil v + \Delta \rceil_r$ lie within distance $3 + \Delta < 4$ eventually. Monotonicity proves (9.3). The interval has width $2\Delta < 3$ when $\Delta < 3/2$, so its ceiling endpoints differ by at most one lattice step. Compare $a_n \geq X$ exactly at the lower candidate when the endpoints differ; this handles equality thresholds without rounding ambiguity. $\square$

Operational boundary. This is an asymptotic bracket theorem. For a supplied finite X , its use as a numerical certificate requires verified onsets for (9.1), logarithmic-error smallness, the derivative bound on $[v - 4, v + 4]$, $\Delta < 1$, and placement of the tested lattice points within the verified range of the model estimate. It also requires a validated computation of the model root and its resulting ceiling bounds. The report does not provide a turnkey interval solver or a single uniform usable onset for all these conditions. The explicit $n \geq 192$ bound in Theorem 4.2 certifies the geometric forward tail condition only; it does not by itself discharge these additional inverse checks.

9.1 The global threshold

An elementary injection strengthens eventual increase to an all-index statement. We record the proof because it removes an otherwise necessary finite-prefix exclusion in threshold inversion.

Proposition 9.2 (Strict increase at every positive index). *For fixed positive integer p , define $A_\varepsilon(n, p) = \lceil w^n \rceil \prod_{j \geq 1} (1 - \varepsilon w^j)^{-\varepsilon j^p}$. Then $A_\varepsilon(n + 1, p) > A_\varepsilon(n, p)$ for every $n \geq 1$, while the values at $n = 0, 1$ are both one. Consequently $A_\varepsilon(n + 1) > A_\varepsilon(n)$ for every $n \geq 1$ in the diagonal family.*

Proof. Represent the j^p colors of a size- j part by words of length p over the alphabet $\{1, \dots, j\}$. In the unrestricted model append one size-one part. This is injective, and a single size- $(n+1)$ part is outside the image, proving strictness.

For the distinct model, a source without a size-one part is sent to itself with the unique size-one part added. Its image has a one. For a source containing a one, and $n \geq 2$, remove that one, select a part of largest size j (breaking a tie by lexicographically largest color word), and replace it by a size- $(j+2)$ part with the same word embedded in the larger alphabet. This is still a valid distinct-colored partition, now of total size $n+1$, with no one. The new part is its unique largest part and is at least two larger than every other part. Recover the source by decreasing that largest size by two, keeping its word, and restoring the one. Thus this map is injective on its image; the two cases have disjoint images. When $n = 1$, send the sole one to a fixed color of a size-two part.

Strictness follows from a single size- $(n+1)$ part whose color word contains the letter $n+1$. It cannot arise from the replacement, whose word uses only letters up to $n-1$, and it has no one. For $n = 1$ there are at least two colors of a size-two part; for $n = 2$ there is no source in the second case, so any single size-three part is outside the image.

Finally, increasing p embeds colors by appending a fixed letter, so $A_\varepsilon(k, p+1) \geq A_\varepsilon(k, p)$ for all k . For $n \geq 1$, combine strict increase in size at exponent $p = n$ with this embedding: $A_\varepsilon(n+1, n+1) \geq A_\varepsilon(n+1, n) > A_\varepsilon(n, n)$. $\square$

The exact adjacent leading-model ratios are

$$\begin{aligned} \frac{L_1(3m+1)}{L_0(3m)} &= 2m(64/9)^m, \\ \frac{L_2(3m+2)}{L_1(3m+1)} &= \frac{2}{m}(81/8)^m, \\ \frac{L_0(3m+3)}{L_2(3m+2)} &= \frac{27}{4(m+1)}(81/8)^m. \end{aligned} \tag{9.5}$$

Each tends to infinity. The equivalents in Theorem 5.1 therefore prove eventual strict increase of both original sequences globally, not only within residue classes, and independently check the eventual behavior already implied by Proposition 9.2. The ordinary threshold inverse, for every threshold where the crossing is defined, is

$$N(X) = \min\{n \geq 0 : a_n \geq X\} = \min_{r=0,1,2} N_r(X). \tag{9.6}$$

Apply the bracket separately to each residue and then minimize the three lower bounds and the three upper bounds. A finite asymptotic series followed by ordinary nearest-integer rounding is not a substitute for this residue-aware procedure.

10 A uniform critical crossover for a varying exponent

The diagonal regime has outer exponent n , far above a second natural scale. To see the transition into an isolated dominant profile, let the exponent vary independently. Write

$$A_\varepsilon(n, \tau) = [w^n] \prod_{j \geq 1} (1 - \varepsilon w^j)^{-\varepsilon j^\tau}, \quad n = 3m + r. \tag{10.1}$$

The combinatorial interpretation assumes positive integer τ . The same analytic theorem holds for real $\tau > 0$ when the product is defined through its formal logarithm. In that extension the distinct model is a signed formal model; positivity of each generalized binomial factor is not asserted. The parameter τ in this section is unrelated to the logarithmic threshold t in the inverse sections.

[write] **5 October 2026.** Part II writes t for this τ (its t is unrelated to the t of Sections 7–9), L_r for $\mathcal{L}_r$, P_ℓ for W_ℓ , x_0 for ξ , (q_r) and (c_r) for κ and g . Its Theorem 15.1 continues this section from bounded z to $0 < z \leq Sm^{3/4}$.

Put

$$z = m^2(8/9)^\tau, \quad a = z^{1/3}, \quad \rho = \frac{5}{\sqrt{32}}, \quad \delta = \frac{2 \log(\sqrt{32}/5)}{\log(9/8)}. \quad (10.2)$$

For the two-parameter family replace n by τ in the exponents, but retain m in the factorials, of the leading models:

$$\mathcal{L}_0 = \frac{3^{m\tau}}{m!}, \quad \mathcal{L}_1 = \frac{3}{2} \frac{4^\tau 3^{(m-1)\tau}}{(m-1)!}, \quad \mathcal{L}_2 = \frac{2^\tau 3^{m\tau}}{m!}. \quad (10.3)$$

Using $H_k = [u^k] \exp(u + u^2)$ from (5.8), define

$$\Psi_0(z) = \sum_{h \geq 0} H_{3h} z^h, \quad \Psi_1(z) = \frac{2}{3} \sum_{h \geq 0} H_{2+3h} z^h, \quad \Psi_2(z) = \sum_{h \geq 0} H_{1+3h} z^h. \quad (10.4)$$

All three are entire functions and $\Psi_r(0) = 1$. Their first terms are

$$\begin{aligned} \Psi_0(z) &= 1 + \frac{7}{6}z + \frac{331}{720}z^2 + O(z^3), \\ \Psi_1(z) &= 1 + \frac{9}{20}z + \frac{1979}{20160}z^2 + O(z^3), \\ \Psi_2(z) &= 1 + \frac{25}{24}z + \frac{1303}{5040}z^2 + O(z^3). \end{aligned}$$

Thus the previously isolated two/four sector ladder is resummed into a nontrivial function when z has positive finite size.

Let $\theta = z d/dz$ and introduce the polynomial operators

$$Q_0(\theta) = Q_2(\theta) = 2\theta^2 - \theta, \quad Q_1(\theta) = 2\theta^2 + \theta, \quad (10.5)$$

$$\begin{aligned} E_0(\theta) &= E_2(\theta) = \frac{1}{6}\theta(\theta-1)(2\theta-1)(6\theta-1), \\ E_1(\theta) &= \frac{1}{6}\theta(2\theta+1)(2\theta-1)(3\theta+1). \end{aligned} \quad (10.6)$$

For real $z \geq 0$, put

$$\Phi_0(z) = \frac{3}{2}z^{3/2}\Psi_1(z), \quad \Phi_1(z) = \frac{2}{3}z^{1/2}\Psi_2(z), \quad \Phi_2(z) = z^{1/2}\Psi_0(z), \quad (10.7)$$

with continuous value zero at $z = 0$.

Theorem 10.1 (Uniform crossover and the first noninteger order). *For each fixed $M > 0$, uniformly in both signs, all residues, and m, τ as $m \rightarrow \infty$ with $0 < z = m^2(8/9)^\tau \leq M$,*

$$\begin{aligned} \frac{A_\varepsilon(3m+r, \tau)}{\mathcal{L}_r} &= \Psi_r(z) - \frac{Q_r(\theta)\Psi_r(z)}{m} + \frac{E_r(\theta)\Psi_r(z)}{m^2} \\ &\quad + \rho^\tau \Phi_r(z) + O_M(m^{-3}). \end{aligned} \quad (10.8)$$

In particular the leading approximation has uniform error $O_M(m^{-1})$, and retaining its first correction gives error $O_M(m^{-2})$. The size-five term equals

$$\rho^\tau \Phi_r(z) = m^{-\delta} z^{\delta/2} \Phi_r(z), \quad 2 < \delta < 3. \quad (10.9)$$

For every fixed positive z this term is nonzero and lies strictly between the second and third algebraic finite-size orders. If it is omitted after the m^{-2} correction, an $o(m^{-\delta})$ remainder is in general impossible.

All constants are independent of the way z approaches zero. Numerically, $2/\log(9/8) \approx 16.9803740$ and $\delta \approx 2.09588823$. At fixed positive z the transition occurs at

$$\tau = \frac{2}{\log(9/8)} \log m - \frac{\log z}{\log(9/8)}. \quad (10.10)$$

For integer τ , a prescribed positive limit z_0 is realized along suitable sequences: take $\tau \rightarrow \infty$ through integers and choose m nearest to $\sqrt{z_0}(9/8)^{\tau/2}$. No claim is made that a fixed rounding rule for τ at every integer m realizes every prescribed limit. If $z \rightarrow z_0 \in [0, \infty)$, the normalized coefficient tends to $\Psi_r(z_0)$.

10.1 Closed forms of the crossover functions

Let $\omega = e^{2\pi i/3}$, and for $q \in \{0, 1, 2\}$ set

$$F_q(a) = \frac{1}{3} \sum_{\ell=0}^2 \omega^{-q\ell} \exp(\omega^\ell a + \omega^{2\ell} a^2). \quad (10.11)$$

The roots-of-unity filter gives

$$\Psi_0(z) = F_0(a), \quad \Psi_1(z) = \frac{2}{3} a^{-2} F_2(a), \quad \Psi_2(z) = a^{-1} F_1(a), \quad a^3 = z. \quad (10.12)$$

The apparent zero singularities are removable. The quotients are independent of the choice of cube root, since $F_q(a) = \sum_{h \geq 0} H_{q+3h} a^{q+3h}$. This identity also proves entireness, for the relevant series converge absolutely at every finite $|a|$. For real positive z , let $B_0 = a + a^2$ and $C_0 = \sqrt{3}(a - a^2)/2$. Then

$$F_q(a) = \frac{1}{3} \left\{ e^{B_0} + 2e^{-B_0/2} \cos(C_0 - 2\pi q/3) \right\}. \quad (10.13)$$

Near zero the power series is preferable for evaluation, since the real formula suffers cancellation for $q = 1, 2$.

10.2 Exact critical-scale atom formula

Isolate $(3, 1)$ in the formal logarithm, and set

$$\xi = (m/3^\tau)^{1/3}, \quad R_{jk} = j/2^{jk/2}, \quad \lambda_{jk} = j^\tau \xi^{jk} = a^{jk/2} R_{jk}^\tau. \quad (10.14)$$

The pure atoms $(2, 1)$ and $(4, 1)$ have activities a, a^2 and $R = 1$. Define the cutoff-normalized falling factorial

$$W_\ell(m) = \begin{cases} (m)_\ell / m^\ell, & 0 \leq \ell \leq m, \\ 0, & \ell > m. \end{cases}$$

For off-core profiles use $D = \sum jkb_{jk}$ as before. Exact extraction of the core gives

$$\frac{A_\varepsilon(3m+r, \tau)}{3^{m\tau}/m!} = \xi^{-r} \sum_{D \equiv r \pmod{3}} W_{(D-r)/3}(m) \prod_{j,k} \frac{(\varepsilon^{k-1} \lambda_{jk}/k)^{b_{jk}}}{b_{jk}!}. \quad (10.15)$$

The zero convention makes the sum effectively finite. With $\kappa = (0, 2, 1)$ and $g = (1, 3/2, 1)$, direct simplification gives

$$\frac{\xi^r \mathcal{L}_r}{3^{m\tau}/m!} = g_r a^{\kappa_r}. \quad (10.16)$$

These identities retain every collision atom and all distinct-model signs; $0 \leq W_\ell \leq 1$ gives an absolute majorant.

10.3 Uniform control of the infinite off-family mass

Call an atom bad if it is neither the core nor either pure atom. Then

$$R_{jk} \leq \rho = 5/\sqrt{32} < 1, \quad (10.17)$$

with equality only at $(5, 1)$. For $k = 1$, the sequence $j2^{-j/2}$ is strictly decreasing from $j = 3$ onwards, since $(j+1)/(j\sqrt{2}) < 1$. After removing $j = 3, 4$, the largest value for $j \geq 5$ is that at 5; $j = 1$ gives $1/\sqrt{2}$. For $k \geq 2$, $R_{jk} \leq j2^{-j} \leq 1/2$.

Fix $M > 0$ and put $B = \max(1, M^{1/6})$. Choose a fixed sufficiently large $T = T(M)$ that $v = B2^{-T/2} < 1$. The convergent majorant

$$S_T(B) = \sum_{\text{bad } (j,k)} \frac{B^{jk} R_{jk}^T}{k} < \infty \quad (10.18)$$

follows by comparison with

$$\sum_{j,k \geq 1} \frac{j^T v^{jk}}{k} = \sum_{j \geq 1} j^T [-\log(1 - v^j)] < \infty.$$

For $\tau \geq T$, therefore, $S_\tau(B) \leq \rho^{\tau-T} S_T(B) = O_M(\rho^\tau)$. The constraint $z \leq M$ ensures $\tau \rightarrow \infty$ uniformly as $m \rightarrow \infty$, so this estimate applies throughout the stated range.

After normalization by a^{κ_r} a degree- D profile carries $a^{D/2-\kappa_r}$. Except for $r = 1, D = 1$, this exponent is nonnegative by the residue condition. Thus $a^{D/2-\kappa_r} \leq B^D$ for $0 < a \leq M^{1/3}$. The total absolute normalized mass of profiles with a bad atom, apart from that single exception, is at most

$$e^{B^2+B^4} (e^{S_\tau(B)} - 1) = O_M(\rho^\tau). \quad (10.19)$$

The exception consists of the single atom $(1, 1)$, with exact normalized contribution

$$\frac{2}{3m} (3/4)^\tau. \quad (10.20)$$

It is also $O_M(\rho^\tau)$. This residue-specific handling is what makes the estimate uniform down to $z = 0$; division of an undifferentiated error by $z^{1/3}$ or $z^{2/3}$ would not suffice.

The identity $\rho^\tau = m^{-\delta} z^{\delta/2}$ is exact. The strict rational inequalities

$$(8/9)^3 < \rho^2 < (8/9)^2$$

reduce to $16384 < 18225$ and $2025 < 2048$, respectively. They prove $2 < \delta < 3$ without using the displayed decimals.

10.4 Two algebraic finite-size corrections

Pure profiles with $b_2 + 2b_4 = k$ have degree $2k$ and aggregate coefficient H_k . The residue condition is $k = \kappa_r + 3h$, so their exact normalized contribution is

$$U_r(m, z) = g_r^{-1} \sum_{h \geq 0} H_{\kappa_r + 3h} z^h W_{2h+1, r=1}(m). \quad (10.21)$$

For nonnegative integers ℓ set

$$s_1(\ell) = \binom{\ell}{2}, \quad e_2(\ell) = \frac{\ell(\ell-1)(\ell-2)(3\ell-1)}{24}.$$

Uniformly in all $\ell \geq 0$ and integer $m \geq 1$,

$$W_\ell(m) = 1 - \frac{s_1(\ell)}{m} + O\left(\frac{(1+\ell)^4}{m^2}\right), \quad (10.22)$$

$$W_\ell(m) = 1 - \frac{s_1(\ell)}{m} + \frac{e_2(\ell)}{m^2} + O\left(\frac{(1+\ell)^6}{m^3}\right). \quad (10.23)$$

For $\ell \leq m+1$, apply the first Bonferroni bounds to the product $\prod_{i=0}^{\ell-1} (1-i/m)$. The respective errors are at most $s_1(\ell)^2/(2m^2)$ and $s_1(\ell)^3/(6m^3)$; e_2 is the sum of the pairwise products of $0, 1, \dots, \ell-1$. This includes $\ell = m+1$, where the product is zero. For $\ell > m+1$, $W_\ell = 0$ and $s_1/m \geq 1$; the absolute values of the displayed approximants are bounded by constant multiples of $(s_1/m)^2$ and $(s_1/m)^3$. This proves the bounds also beyond the exact coefficient cutoff.

For every fixed integer K , the series $\sum_h (1+h)^K H_{q+3h} M^h$ converges, by entireness of $\exp(u+u^2)$. Consequently (10.22) and (10.23) can be summed in (10.21) uniformly on $0 \leq z \leq M$. Substituting $\ell = 2h$ or $2h+1$ gives exactly the operators in (10.5)–(10.6):

$$U_r = \Psi_r - \frac{Q_r(\theta)\Psi_r}{m} + \frac{E_r(\theta)\Psi_r}{m^2} + O_M(m^{-3}). \quad (10.24)$$

Together with (10.19), this already proves the two-term version of the crossover and a remainder $O_M(m^{-\delta})$ after the m^{-2} term.

10.5 The size-five correction and the sharper remainder

After removing (5, 1) as well, every remaining bad atom has $R_{jk} \leq \nu = 3/4$. For $k = 1$ the maximum is at $j = 6$; for $k \geq 2$ the previous bound $1/2$ applies. Repeating the convergent majorant gives activity $O_M(\nu^\tau)$.

Profiles with exactly one (5, 1) atom and otherwise only pure atoms have degree $D = 5 + 2k$. Their residue conditions on k are 2, 1, 0 modulo 3. If W is replaced by one, summing these profiles gives precisely $\rho^\tau \Phi_r(z)$ in (10.7). Restoring W changes the result by $O_M(\rho^\tau/m)$, using $|W_\ell - 1| \leq s_1(\ell)/m$ for all ℓ with the cutoff convention, and the same summable polynomial moments.

All remaining profiles contain another bad atom or at least two (5, 1) atoms. Their absolute normalized mass is $O_M(\nu^\tau + \rho^{2\tau})$, with the exceptional degree-one term (10.20) separately $O(m^{-1}\nu^\tau)$. Hence the entire nonpure contribution is

$$\rho^\tau \Phi_r(z) + O_M\left(\rho^\tau/m + \rho^{2\tau} + \nu^\tau + m^{-1}\nu^\tau\right). \quad (10.25)$$

Set $\eta = 2\log(4/3)/\log(9/8)$, so $\nu^\tau = m^{-\eta}z^{\eta/2}$. The inequality $\nu^2 < (8/9)^3$, equivalently $6561 < 8192$, proves $\eta > 3$. Since $\delta > 2$, every error in (10.25) is $O_M(m^{-3})$, in fact $o_M(m^{-3})$. Combining with (10.24) proves Theorem 10.1. Finally, every H_k is positive, so $\Phi_r(z) > 0$ for fixed $z > 0$, proving that the noninteger-order term is genuinely present.

The diagonal $\tau = n$ has $z \rightarrow 0$ exponentially and is consistent with the isolated-profile limit. This crossover theorem does not replace the sharper diagonal spectral theorem, whose exponentially small relative errors contain information far below the crossover’s uniform algebraic remainder.

[write] **5 October 2026.** On bounded z both this theorem and Part II’s Theorem 15.1 apply. There every displayed correction of Theorem 15.1 is $O(m^{-1})$, so it gives only $A_\varepsilon/(\mathcal{L}_r\Psi_r) = 1 + O(m^{-3/4})$, consistent with, and weaker than, Theorem 10.1, whose m^{-1} , m^{-2} and $m^{-\delta}$ terms lie below Part II’s remainder. On $z \rightarrow \infty$ with $z = O(m^{3/4})$ only Part II applies.

11 Exact computation and reproducibility

The accompanying Python package uses only the standard library. Its arithmetic evidence has three logically separate roles:

1. Exact coefficient checks compare a logarithmic-derivative Euler recurrence, direct product multiplication using binomial coefficients, and multiplication of individual atom exponentials with rational coefficients. The exponent n is fixed throughout each row.
2. Exhaustive phase enumeration implements (5.6)–(5.7), records all eligible atoms, degree bounds and contributing profiles, and compares their aggregates with typed exact fixtures for Tables 1 and 2.
3. Semantic tests run in both ordinary Python and optimized mode. Failure checks are explicit and do not disappear when Python removes `assert` statements. Exact integer and rational fixtures avoid floating-point threshold decisions. Exact crossover checks also cover separate outer exponents, the finite-size polynomial identities, and the rational inequalities that locate the noninteger correction order.

The Euler recurrence itself is worth recording. For an outer exponent n , let $a_0^{(n)} = 1$ and

$$ka_k^{(n)} = \sum_{i=1}^k h_i^{(n,\varepsilon)} a_{k-i}^{(n)}, \quad h_i^{(n,\varepsilon)} = \sum_{d|i} \varepsilon^{i/d-1} d^{n+1}. \quad (11.1)$$

It follows by logarithmic differentiation, and $A_\varepsilon(n) = a_n^{(n)}$. The direct product method instead multiplies the truncated factors with coefficients in (1.2). The atom method multiplies $\exp(\varepsilon^{k-1} j^n x^{jk}/k)$ through degree n . Their agreement is a substantive implementation cross-check because their extraction mechanisms differ.

The mathematical proof is not inferred from finitely many computed values. Exhaustive enumeration proves the finite sector classifications because every excluded branch violates an exact necessary bound; the uniform theorem controls the infinite remainder. The companion does not implement or certify arbitrary finite-threshold inversion.

The archive README gives exact replay commands, the expected file list, and the distinction between code tests and the LaTeX PDF build. The build uses fixed PDF metadata and a fresh

temporary build directory. The archive is formed from an explicit allowlist with fixed ZIP timestamps; auxiliary logs, bytecode, draft notes, and local paths are not deliverables.

[write] **5 October 2026.** The archive README and SHA256SUMS are not shipped; the companion is shipped with the prefix `147-diagonal-` (its README as `147-diagonal-companion-README.md`, its code under `code/`, its certificate under `data/`). The report’s README gives the rerun route from the arrival archive. At the write (Python 3.14.4, Windows, on a fresh extraction) the 56 tests passed in normal and optimized mode, and the regenerated certificate equals `certificate.json` after removal of the carriage returns that Windows text mode adds.

12 Scope of the source and overlap review

The public source review inspected both target entries, both parent arrays, and the fixed-exponent portions of [5]. Exact-ID and bounded topic searches found no correcting diagonal result. This is limited search evidence, not proof that no earlier solution exists. The explicit mathematical claim is a correction of the numerical conjecture in the inspected entries, together with the proved residue sector and inverse structure.

A separate bounded review checked a supplied 178-entry report catalogue, its exclusion manifest, the live series-and-transseries index, and a whole-repository topic/code search. It found no occurrence of either sequence ID or an overlapping diagonal-Euler/integer-break asymptotic topic. The repository main commit at inspection was `bb1cb91b703e3e4d63a39b97470560c64f125cb5`; the code-search index was at its immediate parent, and the intervening change concerned an unrelated computability package. The nearest generic maximum-product and transseries machinery does not state the diagonal conjecture correction or these residue amplitudes. These facts delimit the overlap review; they do not turn generic asymptotic machinery into a new claim.

[write] **5 October 2026.** Commit `bb1cb91b7` exists in the repository history (2 October 2026; it changed only files under `Computability/HilbertTenthProblem/`, as stated). The “supplied 178-entry report catalogue” and its exclusion manifest are not part of the delivery. At the write the only repository files naming A252782 or A270917 are this report’s. The front matter (“The OEIS conjecture, and its refutation”) says which inverse statements are instances of the transseries volume’s apparatus and which are not.

The root-limit proof is elementary. The broader content of the report is the exact defect-phase organization, exhaustive finite classifications, uniform all-depth tail, sign-sensitive separation, and careful smooth-model-to-integer inverse passage. The critical-exponent extension additionally shows how the isolated residue amplitudes become entire crossover functions, with a uniform small-parameter boundary and an explicit noninteger correction.

13 Further questions

Several natural extensions remain open within the scope of this report.

1. **Effective integer inversion.** Derive practical verified onsets for each model estimate and derivative bound, and combine them with an interval-validated root solver. The present asymptotic bracket does not supply this finite-input certificate.
2. **All orders in the critical regime.** Extend the two finite-size corrections to arbitrary order and systematically include nonpure atom families, combining equal decay phases and their interactions with falling-factorial corrections.

3. **Beyond compact crossover windows.** Analyze growing z with quantitative uniformity. A bridge from this regime toward fixed-exponent weighted-partition asymptotics requires further estimates; substitution into either existing regime is not justified here.

[write] **5 October 2026.** *Answered in part* by Part II: Theorem 15.1 and Corollary 15.2 give a uniform expansion, with quantitative remainder $O_S(m^{-3/4})$, for $0 < z \leq Sm^{3/4}$ and every fixed S . The range $z/m^{3/4} \rightarrow \infty$ and the bridge to fixed exponents remain open (Section 27, item 3).

4. **Other integer-weight products.** Determine analogous phase spectra when the weight sequence changes, especially when two integer part sizes compete for dominance or the maximizing size changes.

These are questions for further work, not additional solved claims.

[write] **5 October 2026.** All four questions are carried into Section 27, with Part II's.

Part II

Growing Euler crossover and attenuation inverse

Bundle Report 148, archive `Growing_Euler_Crossover_and_Attenuation_Inverse_Source.zip`, delivered as `Report148.tex` (17 pages). Delivered title block “Growing Euler crossover and attenuation inverse / Uniform moderate defect clouds for diagonal Euler families / Report 148 / 3 October 2026”. Printed as delivered, with `[write]` notes. Delivered Section k is Section $k + 13$ here, and its statement or equation $k.j$ is $(k + 13).j$. Section 27, which closes the report, is added in the write.

Reading conventions of Part II	Elsewhere in the report
t the outer exponent; $\tau_{r,m}(q)$ its inverse value	Part I: τ the exponent; t a logarithmic threshold
$L = \log(9/8)$; L_r the two-parameter amplitudes	Part I: $\mathcal{L}_r$; its $L_r(n)$ are diagonal amplitudes
$\beta = \frac{5}{8}(\delta - 1)$, $\alpha = 5/(4L)$, $\gamma = 8/9$, $\kappa = 1.618\dots$	Part I: β a base, $\alpha = \log 3/3$, $\kappa = (0, 2, 1)$
$P_\ell(m)$; x_0 ; q_r , c_r ; $\rho_{j,k}$; $\mathcal{B}$ the bad atoms	Part I: $W_\ell(m)$; ξ ; κ , g ; R_{jk} ; $\mathcal{B}$ sector bases
$W_\ell = 4w/9$ a variance; $\mathbb{E}$ expectation; Q a compact set	Part I: $W_\ell(m)$; $E_r(\theta)$, $Q_r(\theta)$ operators

Abstract of this Part (as delivered). The two-parameter Euler products associated with A252782 and A270917 have a second crossover when the previously resummed defect activity $z = m^2(8/9)^t$ grows like $m^{3/4}$, with $n = 3m + r$. Relative to the undepleted pure-defect normalizer, the coefficient acquires the uniform attenuation $\exp[-8s^{4/3}/9]$, where $s = z/m^{3/4}$. We prove the first two quarter-power corrections, the residue-dependent term, and a noninteger correction of order $m^{-\beta}$, $\beta \approx 0.684930$, caused by one size-five atom, with remainder $O_S(m^{-3/4})$ for $0 < z \leq Sm^{3/4}$. The proof includes exact conditional-Poisson normalization, global falling-factorial control, the entire growing-radius infinite atom tail, and the vanishing-activity endpoint. A smooth finite model has a unique inverse on a prescribed positive compact window. Every exact real-parameter level crossing in that window is localized by the same expansion, and at least one exists; uniqueness of the exact crossing is not asserted. A new exact standard-library companion checks finite algebra and semantics independently of the analytic proof.

Scope. This report extends the fixed-compact crossover of Report 147. The conditioning, roots-of-unity, moment, and Taylor methods are classical. No global priority claim, all-orders theorem, complete phase diagram, or integer-threshold inversion is claimed here.

14 Definitions and relation to the diagonal problem

For $n \geq 0$, $t > 0$, and $\varepsilon \in \{+1, -1\}$, define

$$A_\varepsilon(n, t) = [x^n] \prod_{j \geq 1} (1 - \varepsilon x^j)^{-\varepsilon j^t}. \quad (14.1)$$

Only $j \leq n$ can contribute to a given coefficient. The parameter t is held fixed throughout that coefficient extraction and throughout any row recurrence. The diagonal specializations $A_+(n, n)$ and $A_-(n, n)$ are the families treated in Report 147 in connection with OEIS A252782 and A270917 [7]. Here t varies separately from n . In the positive moderate window it is logarithmic in m , rather than equal to n . Thus the present result concerns a two-parameter extension of those diagonal sequences.

[write] **5 October 2026.** “Report 147” and [7] are Part I of this report; the unchanged copies under foundation/ are byte-identical to it and are not shipped. Equation (14.1) is Part I’s (10.1) with t for τ , and (14.2) is Part I’s (3.1). The t of this Part is not Part I’s logarithmic threshold t .

For positive integer t , a part of size j has j^t colors. The plus product permits repetition of a colored part; the minus product permits each colored part at most once. The two profile weights are, respectively,

$$\prod_j \binom{j^t + b_j - 1}{b_j}, \quad \prod_j \binom{j^t}{b_j}, \quad \sum_j j b_j = n.$$

For arbitrary real $t > 0$, we instead use the formal logarithm

$$\log \prod_{j \geq 1} (1 - \varepsilon x^j)^{-\varepsilon j^t} = \sum_{j, k \geq 1} \varepsilon^{k-1} \frac{j^t x^{jk}}{k}. \quad (14.2)$$

This definition agrees with the generalized-binomial product. Individual minus factors can have signed coefficients when t is not an integer. No positivity of those factors will be used.

Write $n = 3m + r$, $r \in \{0, 1, 2\}$, and let $m \geq 1$. Put

$$\begin{aligned} L &= \log(9/8), & z &= m^2(8/9)^t, & a &= z^{1/3}, & s &= z/m^{3/4}, \\ \rho &= 5/\sqrt{32}, & \delta &= \frac{2 \log(1/\rho)}{L}, & \beta &= \frac{5}{8}(\delta - 1), & p &= \frac{\delta}{2} + \frac{5}{6}. \end{aligned} \quad (14.3)$$

Here and below $O_S(\cdot)$ means one bound with constants and eventual onset depending only on the fixed positive number S , unless additional subscripts are displayed. All finite choices of sign and residue are included in the same bounds.

The leading residue amplitudes are

$$L_0 = \frac{3^{mt}}{m!}, \quad L_1 = \frac{3}{2} \frac{4^t 3^{(m-1)t}}{(m-1)!}, \quad L_2 = \frac{2^t 3^{mt}}{m!}. \quad (14.4)$$

Although we suppress it in notation, L_r depends on both m and t .

[write] **5 October 2026.** These L_r are Part I’s $\mathcal{L}_r$ of (10.3); on the diagonal $t = n$ they equal Part I’s $L_r(n)$ of (5.1). The letter L alone denotes $\log(9/8)$ in this Part. H_k and Ψ_r are Part I’s (10.4).

Define $H_k = [u^k] \exp(u + u^2)$, so that

$$H_k = \sum_{v=0}^{\lfloor k/2 \rfloor} \frac{1}{(k-2v)!v!}, \quad H_0 = H_1 = 1, \quad H_2 = \frac{3}{2}. \quad (14.5)$$

The undepleted pure-defect functions are

$$\Psi_0(z) = \sum_{h \geq 0} H_{3h} z^h, \quad \Psi_1(z) = \frac{2}{3} \sum_{h \geq 0} H_{2+3h} z^h, \quad \Psi_2(z) = \sum_{h \geq 0} H_{1+3h} z^h. \quad (14.6)$$

Each is entire, positive on $[0, \infty)$, and equals one at zero. Entirety follows, for example, from Cauchy’s bound for $\exp(u + u^2)$ on an arbitrary circle and then letting the circle radius increase.

15 The uniform growing crossover

Theorem 15.1. *Fix $S > 0$. Uniformly for $\varepsilon \in \{+1, -1\}$, $r \in \{0, 1, 2\}$, and positive real t such that $0 < z \leq Sm^{3/4}$, as the positive integer m tends to infinity,*

$$\begin{aligned} \frac{A_\varepsilon(3m+r, t)}{L_r \Psi_r(z)} &= e^{-8s^{4/3}/9} \left[1 - \frac{8}{9} s m^{-1/4} \right. \\ &\quad + \left\{ \frac{4(r-1)}{9} s^{2/3} + \frac{128}{81} s^2 \right\} m^{-1/2} \\ &\quad \left. + s^p m^{-\beta} \right] + O_S(m^{-3/4}). \end{aligned} \quad (15.1)$$

There is no positive lower bound imposed on z or s .

The first correction is independent of sign and residue. Residue dependence first appears at order $m^{-1/2}$. The noninteger correction is positive for both signs, and its exponent satisfies

$$2 < \delta < \frac{11}{5}, \quad \frac{5}{8} < \beta < \frac{3}{4}, \quad \delta = 2.09588822834986\dots, \quad \beta = 0.684930142718664\dots \quad (15.2)$$

These strict inequalities do not depend on rounded decimals. Indeed, $\delta > 2$ is equivalent to $2025 < 2048$, while $\delta < 11/5$ follows from

$$(25/32)^5 > (8/9)^{11}, \quad 306455660244140625 > 288230376151711744.$$

The fractional correction cannot be absorbed in the displayed $O(m^{-3/4})$ remainder at fixed positive s .

[write] 5 October 2026. *A wrong decimal.* The last digits of the decimal of β in (15.2) are wrong: $\beta = \frac{5}{8}(\delta - 1) = 0.684930142718661914\dots$, so the printed $0.684930142718664\dots$ should read $0.684930142718661\dots$ (from the printed, rounded δ one gets 0.6849301427186625). The decimals of δ and of κ in (19.2) are right. Nothing depends on the decimal: the inequalities $2 < \delta < 11/5$ and $5/8 < \beta < 3/4$ are proved by the integer comparisons above, which the write rechecked.

Theorem 15.1 answers in part Part I's further question 3 (Section 13); see the dated note there, and the note at the end of Part I's Section 10 for the overlap of the two theorems on bounded z .

Corollary 15.2. *Uniformly in the same range,*

$$\frac{A_\varepsilon(3m+r, t)}{L_r \Psi_r(z)} = e^{-8s^{4/3}/9} \left(1 - \frac{8}{9} s m^{-1/4} \right) + O_S(m^{-1/2}). \quad (15.3)$$

If $z/m^{3/4} \rightarrow s_0 \in [0, \infty)$, then

$$\frac{A_\varepsilon(3m+r, t)}{L_r \Psi_r(z)} \longrightarrow e^{-8s_0^{4/3}/9}. \quad (15.4)$$

The residue and the sign may vary along the sequence.

The endpoint $s_0 = 0$ includes $z \rightarrow 0$, bounded z , and $z \rightarrow \infty$ with $z = o(m^{3/4})$. At a fixed positive s ,

$$t = t_m(s) = \frac{5}{4L} \log m - \frac{\log s}{L}. \quad (15.5)$$

Each fixed $s > 0$ is realizable even with integer t : let $t \rightarrow \infty$ through integers and take m nearest to $s^{4/5}(9/8)^{4t/5}$. The relative rounding error of m then tends to zero. This observation does not license rounding a continuous inverse in t to an exact integer threshold.

The scale can be anticipated from the pure cloud. Its mean depletion index is asymptotic to $4a^2/3$, while the leading finite-core penalty is $\exp[-\ell^2/(2m)]$. An order-one penalty therefore occurs at $a \asymp m^{1/4}$, equivalently $z \asymp m^{3/4}$. The proof below makes this heuristic uniform and identifies the additional terms.

16 Exact atom normalization and Poisson conditioning

Isolate the logarithmic core atom $(3, 1)$ and set

$$x_0 = (m/3^t)^{1/3} = a^{1/2} 2^{-t/2}, \quad \lambda_{j,k} = j^t x_0^{jk} = a^{jk/2} \left(\frac{j}{2^{jk/2}} \right)^t. \quad (16.1)$$

The equality of the two expressions for x_0 follows exactly from $a^3 = m^2(8/9)^t$. The pure off-core atoms $(2, 1)$ and $(4, 1)$ have activities a and a^2 . Let $\mathcal{B}$ contain every other off-core pair, so it excludes exactly $(2, 1), (3, 1), (4, 1)$.

For a finite off-core profile $b = (b_{j,k})$, write $D = \sum jkb_{j,k}$. When $D \equiv r \pmod{3}$, define $\ell = (D - r)/3$. It is a nonnegative integer. Use the cutoff convention

$$P_\ell(m) = \begin{cases} (m)_\ell / m^\ell, & 0 \leq \ell \leq m, \\ 0, & \ell > m. \end{cases} \quad (16.2)$$

Extracting the remaining core coefficient gives the exact identity

$$\frac{A_\varepsilon(3m + r, t)}{3^{mt}/m!} = x_0^{-r} \sum_{D \equiv r \pmod{3}} P_{(D-r)/3}(m) \prod_{(j,k) \neq (3,1)} \frac{(\varepsilon^{k-1} \lambda_{j,k}/k)^{b_{j,k}}}{b_{j,k}!}. \quad (16.3)$$

Only degrees $D \leq 3m + r$ survive, so this identity is formal without any convergence assumption. Later majorants justify absolutely convergent profile sums after removal of the cutoff.

Put

$$(q_0, q_1, q_2) = (0, 2, 1), \quad (c_0, c_1, c_2) = (1, 3/2, 1), \quad F_q(a) = \sum_{k \equiv q \pmod{3}} H_k a^k. \quad (16.4)$$

Direct substitution in (14.4) yields

$$\frac{x_0^r L_r}{3^{mt}/m!} = c_r a^{q_r}, \quad \Psi_r(a^3) = \frac{F_{q_r}(a)}{c_r a^{q_r}}. \quad (16.5)$$

Thus division of (16.3) by $L_r \Psi_r$ divides its profile sum by $F_{q_r}(a)$, with no further residue factor.

[write] **5 October 2026.** Equations (16.1)–(16.5) restate Part I's (10.14)–(10.16) and (10.12): x_0 is Part I's ξ , P_ℓ its W_ℓ , (q_r) and (c_r) its κ and g , and F_q its F_q .

Let $N_2 \sim \text{Pois}(a)$ and $N_4 \sim \text{Pois}(a^2)$ be independent, and put $K = N_2 + 2N_4$. Then

$$\mathbb{P}(K = k) = e^{-a-a^2} H_k a^k.$$

The pure part of the exact normalized ratio is consequently

$$R_r^{\text{pure}}(m, a) = \mathbb{E} \left[P_{(2K-r)/3}(m) \mid K \equiv q_r \pmod{3} \right]. \quad (16.6)$$

We write $\mathbb{E}_r$ for this conditional expectation. The minimal values of K are 0, 2, 1 for $r = 0, 1, 2$, respectively, so the minimal depletion indices are 0, 1, 0. In particular $P_1 = 1$ makes residue one regular.

17 Uniform control of the residue conditioning

Put $B = a + a^2$ and $\omega = e^{2\pi i/3}$. Filtering the generating function at $1, \omega, \omega^2$ gives

$$F_q(a) = \frac{1}{3} \left\{ e^B + 2e^{-B/2} \cos \left[\frac{\sqrt{3}}{2}(a - a^2) - \frac{2\pi q}{3} \right] \right\}. \quad (17.1)$$

For $a \geq 1$, this implies

$$\mathbb{P}(K \equiv q \pmod{3}) \geq p_* := \frac{1 - 2e^{-3}}{3} > 0. \quad (17.2)$$

For $0 < a \leq 1$ one must first account for the vanishing residue mass. For every fixed integer $h \geq 0$,

$$\mathbb{E}[(1 + K)^h \mid K \equiv q \pmod{3}] \leq \frac{1}{H_q} \sum_{k \equiv q \pmod{3}} (1 + k)^h H_k < \infty. \quad (17.3)$$

This follows by dividing numerator and denominator by a^q , using $F_q(a) \geq H_q a^q$, and then $a^{k-q} \leq 1$. The last series converges because the generating function is entire.

For every nonnegative integer ℓ , including beyond the cutoff,

$$0 \leq 1 - P_\ell(m) \leq \frac{\ell(\ell - 1)}{2m}. \quad (17.4)$$

For $\ell \leq m$, this is the elementary inequality $1 - \prod_i (1 - v_i) \leq \sum_i v_i$ for $0 \leq v_i \leq 1$. For $\ell > m$, its right side is at least one. Hence

$$R_r^{\text{pure}}(m, a) = 1 + O(m^{-1}) \quad (0 < a \leq 1). \quad (17.5)$$

The limit as $a \downarrow 0$ is one for all residues.

For $a \geq 1$, conditioning increases nonnegative expectations by at most p_*^{-1} . The compound-Poisson cumulants of K are $\kappa_j(K) = a + 2^j a^2$. For every fixed integer $h \geq 1$,

$$\mathbb{E}_r |K - (a + 2a^2)|^h \leq C_h a^h, \quad \mathbb{E}_r (1 + K)^h \leq C_h (1 + a^{2h}). \quad (17.6)$$

Indeed, the central moment of even order $2v$ is a sum over partitions with block sizes at least two; each product has at most v cumulants and is $O(a^{2v})$. Other absolute moments follow from Hölder's inequality, and (17.2) transfers the bounds to the conditional law.

Fixed polynomial moments also become equidistributed among residues. Differentiating the filter gives, for a polynomial of fixed degree h , a conditional-minus-unconditional error bounded by a constant times

$$(1 + a)^{2h} e^{-3B/2} \quad (a \geq 1). \quad (17.7)$$

The reason is that every fixed Euler derivative $(u d/du)^h$ of $\exp(au + a^2 u^2)$ at $u = \omega, \omega^2$ equals that exponential times a polynomial in a , while its modulus is $e^{-B/2}$. Division by the residue normalizer costs at most $p_*^{-1} e^{-B}$. Additional fixed polynomials in a preserve exponential decay. These observations supply all moment replacements made below, without applying a large- a estimate at small a .

18 Pure cloud depletion with a global remainder

Proposition 18.1. *For every fixed $A > 0$, uniformly for $0 < a \leq Am^{1/4}$ and all residues,*

$$R_r^{\text{pure}}(m, a) = e^{-8a^4/(9m)} \left[1 - \frac{8a^3}{9m} + \frac{4(r-1)a^2}{9m} + \frac{128a^6}{81m^2} \right] + O_A(m^{-3/4}). \quad (18.1)$$

Proof. For $0 < a \leq 1$, (17.5) suffices: the displayed approximation also equals $1 + O(m^{-1})$. Suppose therefore that $1 \leq a \leq Am^{1/4}$. Define

$$f_m(x) = \exp \left[-\frac{x^2 - x}{2m} - \frac{x^3}{6m^2} \right].$$

On the nonnegative integer support, $0 < f_m \leq 1$. For $0 \leq \ell \leq m/2$, expansion of $\log(1 - i/m)$ and summation over $0 \leq i < \ell$ give

$$|\log P_\ell - \log f_m(\ell)| \leq C \left(\frac{\ell^2}{m^2} + \frac{\ell^4}{m^3} \right). \quad (18.2)$$

Both logarithms are nonpositive, so exponentiation is 1-Lipschitz there. On $\ell > m/2$, simply use $|P_\ell - f_m(\ell)| \leq 1$ and the fourth moment Markov bound. This covers $\ell > m$ as well. By (17.6),

$$\begin{aligned} \mathbb{E}_r |P_\ell - f_m(\ell)| &\leq C \left\{ \frac{1+a^4}{m^2} + \frac{1+a^8}{m^3} + \frac{1+a^8}{m^4} \right\} \\ &= O_A(m^{-1}), \quad \ell = (2K - r)/3. \end{aligned} \quad (18.3)$$

The cutoff has thus been handled globally rather than by an unquantified “typical profile” restriction.

Set $d = 4a^2/3$ and $X = \ell - d$. Then $f_m(\ell) = e^{-d^2/(2m)} e^V$, where exactly

$$V = -\frac{dX}{m} - \frac{X^2}{2m} + \frac{d+X}{2m} - \frac{(d+X)^3}{6m^2}. \quad (18.4)$$

The displacement of d from the unconditioned mean of ℓ is $(2a - r)/3 = O(a)$, so (17.6) implies $\mathbb{E}_r |X|^h \leq C_h a^h$ for every fixed h . Each monomial of (18.4) has L^3 norm bounded by a constant times one of

$$a^3/m, a^2/m, a/m, a^6/m^2, a^5/m^2, a^4/m^2, a^3/m^2.$$

It follows that $\|V\|_3 = O_A(m^{-1/4})$. Moreover, $V \leq d^2/(2m) \leq 8A^4/9$, because $\log f_m(\ell) \leq 0$ on the support. The exponential Taylor remainder obeys $|e^v - 1 - v - v^2/2| \leq C_A |v|^3$ for all $v \leq 8A^4/9$, even with an unbounded negative tail. Therefore

$$\mathbb{E}_r e^V = 1 + \mathbb{E}_r V + \frac{1}{2} \mathbb{E}_r V^2 + O_A(m^{-3/4}). \quad (18.5)$$

The required unconditional moments are

$$\mathbb{E} X = \frac{2a - r}{3}, \quad \mathbb{E} X^2 = \frac{20a^2 + 4(1 - r)a + r^2}{9}. \quad (18.6)$$

Equation (17.7) permits their conditional replacements: each error is a polynomial in a times $e^{-3B/2}$ and is multiplied by at least $1/m$. The total replacement error is $O(m^{-1})$. Expanding (18.4) and retaining terms larger than the target error now gives

$$\mathbb{E}_r V = -\frac{8a^3}{9m} + \frac{4(r-1)a^2}{9m} - \frac{32a^6}{81m^2} + O_A(m^{-3/4}). \quad (18.7)$$

For example, all a/m terms are $O_A(m^{-3/4})$, and all terms of size $(1 + a^5)/m^2$ are also $O_A(m^{-3/4})$.

For the quadratic term, write $V = -dX/m + W$. For every fixed moment order the norm of W is $O_A(m^{-1/2})$, whereas that of dX/m is $O_A(m^{-1/4})$. Hölder's inequality bounds the cross term and W^2 by $O_A(m^{-3/4})$. Using (18.6) yields

$$\frac{1}{2}\mathbb{E}_r V^2 = \frac{d^2}{2m^2}\mathbb{E} X^2 + O_A(m^{-3/4}) = \frac{160a^6}{81m^2} + O_A(m^{-3/4}). \quad (18.8)$$

Combining (18.3), (18.5), (18.7), and (18.8) proves the proposition. $\square$

Lemma 18.2. *Fix $A, B_0 > 0$. Suppose $|b| \leq B_0$ and a residue condition $K \equiv q \pmod{3}$ makes $(2K + b)/3$ a nonnegative integer on its support. Uniformly for $1 \leq a \leq Am^{1/4}$,*

$$\mathbb{E} \left[P_{(2K+b)/3}(m) \mid K \equiv q \pmod{3} \right] = e^{-8a^4/(9m)} + O_{A,B_0}(m^{-1/4}). \quad (18.9)$$

Proof. The change in b changes X by a bounded constant. The moment bounds, global approximation, and upper bound on V remain uniform. Retaining only $\mathbb{E}|V| = O_{A,B_0}(m^{-1/4})$ in the preceding proof proves the claim. Only the three shifts $b = 5 - r$ will be used. $\square$

19 The entire growing radius bad atom sum

Define

$$\mathcal{S}_{\text{bad}} = \sum_{(j,k) \in \mathcal{B}} \frac{\lambda_{j,k}}{k}, \quad \lambda_5 = \lambda_{5,1} = a^{5/2} \rho^t. \quad (19.1)$$

The next bound controls both infinite indices, including all collisions of the core and pure sizes. It does not transfer constants from a fixed-radius result.

Proposition 19.1. *Uniformly for $0 < a \leq S^{1/3}m^{1/4}$ with the exact relation $a^3 = m^2(8/9)^t$,*

$$\mathcal{S}_{\text{bad}} = \lambda_5 \{1 + O_S(m^{-\kappa})\}, \quad \kappa = \frac{5}{4L} \log \frac{5\sqrt{2}}{6} - \frac{1}{8} = 1.61816310250741 \dots \quad (19.2)$$

In particular,

$$\lambda_5 = m^{-\delta} a^{(3\delta+5)/2} = m^{-\beta} s^p, \quad \mathcal{S}_{\text{bad}} = O_S(m^{-\beta}). \quad (19.3)$$

Proof. To avoid confusing a radius with the inverse target used later, denote

$$\xi = x_0 = a^{1/2} 2^{-t/2}, \quad R = a^{1/2} \left(\frac{6}{5\sqrt{2}} \right)^t, \quad T = a(3/4)^t.$$

Since $t = (2 \log m - 3 \log a)/L$, the upper endpoint for a gives

$$\xi \leq C_\xi(S) m^{-\gamma_\xi}, \quad R \leq C_R(S) m^{-\kappa}, \quad T \leq C_T(S) m^{-\gamma_T}, \quad (19.4)$$

where

$$\begin{aligned} C_\xi(S) &= S^{1/6 + \log 2/(2L)}, & \gamma_\xi &= \frac{5 \log 2}{8L} - \frac{1}{8}, \\ C_R(S) &= S^{1/6 + \log(5\sqrt{2}/6)/L}, & C_T(S) &= S^{1/3 + \log(4/3)/L}, \\ \gamma_T &= \frac{5}{4L} \log(4/3) - \frac{1}{4}. \end{aligned} \quad (19.5)$$

All three exponents are positive. Eventually $\xi, R, T \leq 1/2$. For every $j \geq 5$,

$$\frac{\lambda_{j+1,1}}{\lambda_{j,1}} = \xi \left(\frac{j+1}{j} \right)^t \leq R,$$

whence the entire single-atom degree tail satisfies

$$\lambda_5 \leq \sum_{j \geq 5} \lambda_{j,1} \leq \frac{\lambda_5}{1-R}. \quad (19.6)$$

For the collision family, sum all $k \geq 2$ first:

$$\sum_{k \geq 2} \frac{j^t \xi^{jk}}{k} \leq \frac{j^t \xi^{2j}}{2(1-\xi^j)} \leq \frac{j^t \xi^{2j}}{2(1-\xi)}.$$

Put $b_j = j^t \xi^{2j}$. For $j \geq 2$, $b_{j+1}/b_j \leq \xi^2(3/2)^t = T$, while $b_1 = a2^{-t}$ and $b_2 = a^2 2^{-t}$. Thus

$$\sum_{j \geq 1, k \geq 2} \frac{\lambda_{j,k}}{k} \leq \frac{b_1 + b_2/(1-T)}{2(1-\xi)} \leq b_1 + 2b_2. \quad (19.7)$$

This also establishes convergence of the infinite activity sum. The remaining $k = 1$ bad atom is $(1, 1)$, so

$$\lambda_5 \leq \mathcal{S}_{\text{bad}} \leq \frac{\lambda_5}{1-R} + \lambda_{1,1} + b_1 + 2b_2. \quad (19.8)$$

The comparison with λ_5 must remain valid as $a \downarrow 0$. Write $h = \log(5/4)$, $d_0 = \log(5/(2\sqrt{2}))$, and $\alpha = 5/(4L)$. Exactly,

$$\begin{aligned} \frac{\lambda_{1,1}}{\lambda_5} &= m^{-2h/L} a^{3h/L-2}, \\ \frac{b_1}{\lambda_5} &= m^{-2d_0/L} a^{3d_0/L-3/2}, \\ \frac{b_2}{\lambda_5} &= m^{-2d_0/L} a^{3d_0/L-1/2}. \end{aligned} \quad (19.9)$$

Every displayed power of a is positive: the first assertion is $125 > 81$, the second follows from $25/8 > 9/8$, and the third is stronger. Maximization at the allowed upper endpoint therefore gives

$$\begin{aligned} \lambda_{1,1}/\lambda_5 &\leq S^{h/L-2/3} m^{-\gamma_1}, & \gamma_1 &= \kappa + 5/4, \\ b_1/\lambda_5 &\leq S^{d_0/L-1/2} m^{-\gamma_{c1}}, & \gamma_{c1} &= \alpha d_0 + 3/8, \\ b_2/\lambda_5 &\leq S^{d_0/L-1/6} m^{-\gamma_{c2}}, & \gamma_{c2} &= \alpha d_0 + 1/8. \end{aligned} \quad (19.10)$$

Here

$$\gamma_{c2} - \kappa = \alpha \log(3/2) + 1/4 > 0, \quad \gamma_{c1} = \gamma_{c2} + 1/4.$$

Using $(1-R)^{-1} \leq 1 + 2R$ now proves (19.2). Finally, substitution of the exact relation for t gives (19.3). $\square$

For an operational bound, once (19.4) is at most $1/2$ in each component, a permissible upper bound for $\mathcal{S}_{\text{bad}}/\lambda_5 - 1$ is

$$2C_R(S)m^{-\kappa} + S^{h/L-2/3}m^{-\gamma_1} + S^{d_0/L-1/2}m^{-\gamma_{c1}} + 2S^{d_0/L-1/6}m^{-\gamma_{c2}}. \quad (19.11)$$

A sufficient onset for the radius inequalities is the maximum of one and $(2C_\xi)^{1/\gamma_\xi}$, $(2C_R)^{1/\kappa}$, $(2C_T)^{1/\gamma_T}$, with a further strict increase if needed to ensure $t > 0$. The forward theorem as a whole does not provide an explicit universal finite-input onset, since its moment constants have not all been optimized.

Notice that $\kappa > 1/2$ already follows from $(5\sqrt{2}/6)^2 = 25/18 > 9/8$. The order of the absolute activity bound is sharp at $z = Sm^{3/4}$, since the single positive activity is exactly $\lambda_5 = S^p m^{-\beta}$.

20 The size five contribution and the zero endpoint

First suppose $a \geq 1$. Dropping residue constraints and using $P_\ell \leq 1$ bounds the unsigned relative mass of profiles containing at least one bad atom by

$$p_*^{-1}(e^{\mathcal{S}_{\text{bad}}} - 1) = O_S(m^{-\beta}). \quad (20.1)$$

This is enough for (15.3), since $\beta > 1/2$, but not enough for (15.1). We must isolate exactly one size-five atom.

A profile with one $(5, 1)$ atom and otherwise pure atoms has degree $5 + 2K$. Its required residue is

$$q'_r = (q_r + 2) \bmod 3, \quad (q'_0, q'_1, q'_2) = (2, 1, 0). \quad (20.2)$$

The exact single-five contribution to the normalized ratio is

$$V_r = \lambda_5 \frac{F_{q'_r}(a)}{F_{q_r}(a)} \mathbb{E} \left[P_{(5+2K-r)/3}(m) \mid K \equiv q'_r \pmod{3} \right]. \quad (20.3)$$

The minimal indices in the three residues are 3, 2, 1. This term is positive for both signs because $k = 1$. Equations (17.1) and (18.9) give

$$\frac{F_{q'_r}}{F_{q_r}} = 1 + O(e^{-3B/2}), \quad \mathbb{E}[P_{(5+2K-r)/3} \mid K \equiv q'_r] = e^{-8a^4/(9m)} + O_S(m^{-1/4}).$$

Moreover,

$$\lambda_5 e^{-3B/2} = m^{-\delta} a^{(3\delta+5)/2} e^{-3B/2} = O(m^{-\delta}),$$

uniformly for $a \geq 1$. Thus

$$V_r = \lambda_5 e^{-8a^4/(9m)} + O_S(m^{-\beta-1/4} + m^{-\delta}). \quad (20.4)$$

Set $T_{\text{bad}} = \mathcal{S}_{\text{bad}} - \lambda_5 \geq 0$. All remaining nonpure profiles contain at least two size-five atoms or another bad atom. Their total relative absolute mass is at most

$$\begin{aligned} p_*^{-1} \{ e^{\lambda_5} - 1 - \lambda_5 + e^{\lambda_5} (e^{T_{\text{bad}}} - 1) \} \\ = O_S(\lambda_5^2 + T_{\text{bad}}) = O_S(m^{-2\beta} + m^{-\beta-\kappa}). \end{aligned} \quad (20.5)$$

Every error exponent in (20.4) and (20.5) is strictly greater than $3/4$: $\beta + 1/4 > 7/8$, $\delta > 2$, $2\beta > 5/4$, and $\beta + \kappa > 9/8$. These are absolute estimates and hence remain valid for the signed t -minus model.

20.1 Why small activity needs a separate argument

[write] **5 October 2026.** This subsection repeats, in Part II's normalization, the small-activity argument of Part I's Section 10: the bad-ratio bound (10.17), the fixed majorant (10.18) and the exceptional term (10.20), which equals (20.7).

For $0 < a \leq 1$, the normalizers F_1 and F_2 vanish with a . Dividing a residue-free error by them would not establish uniformity. Instead, take a profile of degree $D \equiv r \pmod{3}$. After division by $F_{q_r}(a) \geq c_r a^{q_r}$ its activity has a residual factor $a^{D/2 - q_r}$. This power is nonnegative in every admissible profile except $r = 1, D = 1$, the unique profile consisting of one $(1, 1)$ atom.

Exclude that profile temporarily. Every bad ratio $\rho_{j,k} = j/2^{j^{k/2}}$ is at most $\rho = 5/\sqrt{32} < 1$. For $k = 1$, $j/2^{j/2}$ decreases from $j = 3$ onward; the remaining $j = 1$ term is smaller. For $k \geq 2$, $\rho_{j,k} \leq j/2^j \leq 1/2$. Fix a sufficiently large $T_0 > 0$. Then

$$\sum_{\mathcal{B}} \frac{\rho_{j,k}^t}{k} \leq \rho^{t-T_0} \sum_{\mathcal{B}} \frac{\rho_{j,k}^{T_0}}{k} = O(\rho^t). \quad (20.6)$$

The fixed sum converges, since the larger sum over all pairs equals

$$\sum_{j \geq 1} j^{T_0} [-\log(1 - 2^{-T_0 j/2})] < \infty.$$

After dropping the nonnegative residual powers of a , unrestricted pure profiles contribute at most e^2 . Exponentiating the convergent bad-atom majorant therefore bounds all nonexceptional profiles with a bad atom by $O(\rho^t)$ relative to their residue normalizer.

The excluded degree-one profile has relative mass at most

$$\frac{2}{3} a^{-3/2} 2^{-t/2} = \frac{2}{3m} (3/4)^t. \quad (20.7)$$

Since $a \leq 1$ implies $t \geq 2 \log m/L$,

$$\rho^t = m^{-\delta} a^{3\delta/2} \leq m^{-\delta}, \quad (3/4)^t \leq m^{-\eta}, \quad \eta = 2 \log(4/3)/L > 2.$$

The exceptional term is smaller than $O(m^{-\delta})$, because $1 + \eta > 3 > \delta$. Thus the full nonpure contribution is $O(m^{-\delta})$ uniformly at this endpoint. Together with (17.5), the exact ratio is $1 + O(m^{-1})$ there, and so is the right side of (15.1). This is stronger than needed.

20.2 Completion of the forward theorem

For $a \geq 1$, Proposition 18.1, (20.4), and (20.5) give

$$\frac{A_\varepsilon}{L_r \Psi_r} = e^{-8a^4/(9m)} \left[1 - \frac{8a^3}{9m} + \frac{4(r-1)a^2}{9m} + \frac{128a^6}{81m^2} + \lambda_5 \right] + O_S(m^{-3/4}).$$

Use

$$\frac{a^4}{m} = s^{4/3}, \quad \frac{a^3}{m} = sm^{-1/4}, \quad \frac{a^2}{m} = s^{2/3} m^{-1/2}, \quad \frac{a^6}{m^2} = s^2 m^{-1/2}, \quad \lambda_5 = s^p m^{-\beta}.$$

The small- a argument covers the entire remaining range, proving Theorem 15.1. Dropping terms of bounded coefficient proves Corollary 15.2.

21 The attenuation inverse on a prescribed window

The inverse result concerns a real exponent parameter. Its unique root belongs to a smooth finite model; the exact ratio is only shown to have one or more crossings in a narrow bracket.

Set $\gamma = 8/9$, $\alpha = 5/(4L)$, $u = m^{-1/4}$, and $v = m^{-\beta}$. For $s > 0$, define $t_m(s)$ by (15.5) and

$$R_{\varepsilon,r,m}(s) = \frac{A_\varepsilon(3m+r, t_m(s))}{L_r(m, t_m(s))\Psi_r(sm^{3/4})}. \quad (21.1)$$

On any fixed compact interval $I \subset (0, \infty)$, the forward theorem implies

$$R_{\varepsilon,r,m}(s) = \mathcal{M}_{r,m}(s) + O_I(u^3), \quad (21.2)$$

where the positive finite model is

$$\mathcal{M}_{r,m}(s) = \exp \left[-\gamma s^{4/3} - \gamma s u + \left\{ \frac{4(r-1)}{9} s^{2/3} + \frac{32}{27} s^2 \right\} u^2 + s^p v \right]. \quad (21.3)$$

To check the changed quadratic coefficient, taking the logarithm of the bracket in (15.1) subtracts $\gamma^2 s^2/2 = 32s^2/81$ from $128s^2/81$, leaving $32s^2/27$. The mixed errors uv and v^2 are $O(u^3)$ because $\beta + 1/4 > 3/4$ and $2\beta > 3/4$. On I the leading factor stays bounded away from zero; for large m the exact ratio is positive by the forward theorem. This justifies the logarithm despite possible signed formal coefficients at other real parameter values.

Let Q be a fixed compact subset of $(0, 1)$, and set

$$s_0(q) = \left(\frac{-\log q}{\gamma} \right)^{3/4}. \quad (21.4)$$

Choose a fixed compact interval $I = [b_-, b_+] \subset (0, \infty)$ whose interior contains every $s_0(q)$ for $q \in Q$. For example, if $Q \subset [q_-, q_+] \subset (0, 1)$, take $I = [s_0(q_+)/2, 2s_0(q_-)]$.

Theorem 21.1. *Uniformly for $q \in Q$ and all residues, for sufficiently large m the model $\mathcal{M}_{r,m}$ is strictly decreasing on I , and $\mathcal{M}_{r,m}(\sigma) = q$ has a unique solution $\sigma_{r,m}(q)$ in I . Writing $s_0 = s_0(q)$, its expansion is*

$$\begin{aligned} \sigma_{r,m}(q) &= s_0 - \frac{3}{4} s_0^{2/3} u \\ &\quad + \left\{ \frac{3(4r+1)}{32} s_0^{1/3} + s_0^{5/3} \right\} u^2 \\ &\quad + \frac{27}{32} s_0^{(\delta+1)/2} v + O_{Q,I}(u^3). \end{aligned} \quad (21.5)$$

Its exponent parameter $\tau_{r,m}(q) = t_m(\sigma_{r,m}(q))$ satisfies

$$\begin{aligned} \tau_{r,m}(q) &= \frac{5}{4L} \log m - \frac{\log s_0}{L} + \frac{3s_0^{-1/3}}{4L} u \\ &\quad - \frac{1}{L} \left\{ \frac{3(2r-1)}{16} s_0^{-2/3} + s_0^{2/3} \right\} u^2 \\ &\quad - \frac{27}{32L} s_0^{(\delta-1)/2} v + O_{Q,I}(u^3). \end{aligned} \quad (21.6)$$

Proof. Write the logarithm of the model as $F(s) + G(s)u + H_r(s)u^2 + J(s)v$, where

$$F = -\gamma s^{4/3}, \quad G = -\gamma s, \quad H_r = \frac{4(r-1)}{9} s^{2/3} + \frac{32}{27} s^2, \quad J = s^p.$$

Its derivative converges uniformly on I to $F'(s) = -(4\gamma/3)s^{1/3} < 0$. Both the logarithmic derivative and the model derivative are eventually bounded above by a negative constant. The limiting roots lie a positive distance from the endpoints, so uniform endpoint straddling and the intermediate value theorem give existence and uniqueness of the model root in I .

Substitute $s = s_0 + c_1 u + c_2 u^2 + c_b v$. Taylor expansion, using $F(s_0) = \log q$, gives the three coefficient equations

$$\begin{aligned} F'c_1 + G &= 0, \\ F'c_2 + \frac{1}{2}F''c_1^2 + G'c_1 + H_r &= 0, \\ F'c_b + J &= 0, \end{aligned} \tag{21.7}$$

all evaluated at s_0 . Solving yields

$$c_1 = -\frac{3}{4}s_0^{2/3}, \quad c_2 = \frac{3(4r+1)}{32}s_0^{1/3} + s_0^{5/3}, \quad c_b = \frac{27}{32}s_0^{(\delta+1)/2}. \tag{21.8}$$

All derivatives are bounded on a fixed positive compact neighborhood of the limiting roots. Pure cubic and mixed terms are $O_{Q,I}(u^3)$, so substitution leaves a logarithmic-model residual of that order. The derivative lower bound and the mean-value theorem convert it into a root error, proving (21.5).

Finally,

$$\log \sigma = \log s_0 + \frac{c_1}{s_0}u + \left(\frac{c_2}{s_0} - \frac{c_1^2}{2s_0^2} \right)u^2 + \frac{c_b}{s_0}v + O_{Q,I}(u^3),$$

with

$$\frac{c_2}{s_0} - \frac{c_1^2}{2s_0^2} = \frac{3(2r-1)}{16}s_0^{-2/3} + s_0^{2/3}.$$

Substitution into $t_m(\sigma) = \alpha \log m - (\log \sigma)/L$ proves (21.6). $\square$

22 What the exact level crossings inherit

Let s_{app} and t_{app} denote the displayed expressions in (21.5) and (21.6), with their error terms omitted.

Theorem 22.1. *For the compact sets Q, I above there exist constants $C, D > 0$ and a uniform eventual onset, valid for both signs and all residues, such that*

$$R_{\varepsilon, r, m}(s_{\text{app}} - Cu^3) > q, \quad R_{\varepsilon, r, m}(s_{\text{app}} + Cu^3) < q. \tag{22.1}$$

At least one exact real-parameter level- q crossing occurs between these endpoints. Every exact crossing $s \in I$ satisfies

$$|s - s_{\text{app}}| \leq Cu^3. \tag{22.2}$$

Put $J_m = t_m(I)$ and $s(t) = m^{5/4}(8/9)^t$. Then

$$R_{\varepsilon, r, m}(s(t_{\text{app}} - Du^3)) < q, \quad R_{\varepsilon, r, m}(s(t_{\text{app}} + Du^3)) > q. \tag{22.3}$$

Every exact real- t crossing in J_m lies within $Dm^{-3/4}$ of t_{app} , and at least one exists in that bracket.

Proof. For fixed n , the formal coefficient in (14.1) is a finite sum of finite products of the continuous functions j^t . Its normalizer is positive. Thus the exact ratio is continuous in real t , and hence in s on I for all sufficiently large m . This argument uses no generalized-binomial nonnegativity.

Let K_0 bound the uniform error in (21.2), and let K_1 bound $|\mathcal{M}(s_{\text{app}}) - q|/u^3$. Choose $c > 0$ so that $-\mathcal{M}'(s) \geq c$ throughout I . All constants can be chosen uniformly. For $C > (K_0 + K_1)/c$, the mean-value theorem gives

$$R(s_{\text{app}} - Cu^3) - q \geq (cC - K_1 - K_0)u^3 > 0,$$

and the analogous strictly negative bound at the right endpoint. These points lie inside I eventually. Continuity proves existence.

If $R(s_*) = q$ at any $s_* \in I$, then $|\mathcal{M}(s_*) - q| \leq K_0 u^3$. Since $\mathcal{M}$ has its unique root σ and derivative magnitude at least c , $|s_* - \sigma| \leq K_0 u^3/c$. Combining this with (21.5) proves (22.2), after enlarging C . The same reasoning classifies the sign of $R - q$ outside the bracket throughout I .

The map t_m is decreasing with derivative $-1/(Ls)$, bounded above and below in magnitude on I . Also $t_m(s_{\text{app}}) = t_{\text{app}} + O_{Q,I}(u^3)$ by the logarithmic expansion. Transforming the bracket, reversing its orientation, and enlarging its constant proves (22.3) and the t localization. $\square$

These conclusions are uniform asymptotic residual brackets, with constants and eventual onsets; they are not a numerical certification at a prescribed finite m . A uniform approximation of function values does not imply approximation of derivatives. No monotonicity or uniqueness of the exact ratio has been proved. Nor does this identify a first crossing over all $t > 0$ or exclude crossings outside J_m . Integer values of t usually do not attain a prescribed q exactly. A unit change in t multiplies s by $8/9$, so a continuous inverse cannot simply be rounded while retaining the $O(m^{-3/4})$ accuracy.

23 An independent global factorial derivation

For completeness we give a second route to the pure-cloud coefficients. Besides checking the algebra, this route records an all- ℓ factorial remainder without using a cutoff-tail expectation. Write

$$g_m(x) = e^{-x(x-1)/(2m)}, \quad y_\ell = \frac{\ell(\ell-1)(2\ell-1)}{12m^2}.$$

Lemma 23.1. *For all integers $m \geq 1$, $\ell \geq 0$, with (16.2),*

$$|P_\ell(m) - g_m(\ell)(1 - y_\ell)| \leq C \left(\frac{\ell^4}{m^3} + \frac{\ell^6}{m^4} \right). \quad (23.1)$$

Proof. For $\ell \leq m/2$, summing the logarithmic series gives

$$\log P_\ell = -\frac{\ell(\ell-1)}{2m} - y_\ell - \mathcal{R}_\ell, \quad 0 \leq \mathcal{R}_\ell \leq C\ell^4/m^3.$$

For $y, \mathcal{R} \geq 0$, $|e^{-y-\mathcal{R}} - (1-y)| \leq y^2/2 + \mathcal{R}$, which proves the bound there. For $\ell > m/2$, use $|P_\ell - g_m(\ell)(1 - y_\ell)| \leq 2 + y_\ell$. If $\ell \geq 2$, a universal multiple of ℓ^4/m^3 dominates both two and y_ℓ because $m < 2\ell$; the cases $\ell = 0, 1$ are exact. This includes the entire region $\ell > m$. $\square$

Put $\mu = a + 2a^2$, $w = a + 4a^2$, and

$$\mu_\ell = (2\mu - r)/3, \quad W_\ell = 4w/9, \quad \Delta = \ell - \mu_\ell.$$

For $a \geq 1$, the moments and residue corrections established above give $\mathbb{E}_r|\Delta|^h \leq C_h a^h$ for fixed h , and

$$\mathbb{E}_r \Delta = O((a + a^2)e^{-3B/2}), \quad \mathbb{E}_r \Delta^2 - W_\ell = O((a + a^4)e^{-3B/2}).$$

Averaging (23.1) gives a remainder

$$O\left((1 + a^8)/m^3 + (1 + a^{12})/m^4\right) = O_A(m^{-1}).$$

Since $g_m(x) = e^{1/(8m)} e^{-(x-1/2)^2/(2m)}$, its fixed derivatives satisfy $\sup_x |g_m^{(j)}(x)| \leq C_j m^{-j/2}$. Taylor expansion around μ_ℓ gives

$$\mathbb{E}_r g_m(\ell) = g_m(\mu_\ell) + \frac{1}{2} g_m''(\mu_\ell) W_\ell + O\left(m^{-1} + (1 + a^3)m^{-3/2}\right). \quad (23.2)$$

For the first derivative term, the exact formula $g'_m(\mu_\ell) = -(\mu_\ell - 1/2)g_m(\mu_\ell)/m$ and the exponentially small residue factor give $O(m^{-1})$. The second moment replacement also costs $O(m^{-1})$.

For $h_m(x) = x^3 g_m(x)$, $\sup_{x \geq -2/3} |h'_m(x)| \leq Cm$, by Gaussian scaling. Since $\mu_\ell \geq -2/3$ and $\ell \geq 0$, the mean-value theorem gives

$$\mathbb{E}_r[g_m(\ell)y_\ell] = \frac{g_m(\mu_\ell)\mu_\ell^3}{6m^2} + O\left((1 + a)/m + (1 + a^4)/m^2\right). \quad (23.3)$$

Together these formulas yield

$$\mathbb{E}_r P_\ell = g_m(\mu_\ell) \left[1 - \frac{W_\ell}{2m} + \frac{(\mu_\ell - 1/2)^2 W_\ell}{2m^2} - \frac{\mu_\ell^3}{6m^2} \right] + O_A(m^{-3/4}). \quad (23.4)$$

The bracket is $1 + O_A(m^{-1/2})$, and $g_m(\mu_\ell)$ is bounded below by a positive A -dependent constant in the allowed range. Its logarithm may therefore be taken with the same error order. The exact polynomial identity

$$\begin{aligned} -\frac{\mu_\ell(\mu_\ell - 1) + W_\ell}{2m} &= -\frac{8a^4}{9m} - \frac{8a^3}{9m} + \frac{4(r-1)a^2}{9m} \\ &\quad + \frac{(2r+1)a}{9m} - \frac{r(r+3)}{18m} \end{aligned} \quad (23.5)$$

and the leading-degree computation

$$\frac{(\mu_\ell - 1/2)^2 W_\ell}{2m^2} - \frac{\mu_\ell^3}{6m^2} = \frac{32a^6}{27m^2} + O((1 + a^5)/m^2) \quad (23.6)$$

show that

$$\log R_r^{\text{pure}} = -\frac{8a^4}{9m} - \frac{8a^3}{9m} + \frac{4(r-1)a^2}{9m} + \frac{32a^6}{27m^2} + O_A(m^{-3/4}). \quad (23.7)$$

Exponentiation adds $32a^6/(81m^2)$, recovering 128/81 in (18.1). The small- a argument remains (17.5); no Gaussian large-activity approximation is needed at that endpoint.

24 Exact companion and reproducibility

The source archive contains a newly written Python standard-library companion, the report source, deterministic local build and archive scripts, and unchanged PDF and TeX copies of Report 147 under `foundation/`. It does not include or depend on that report’s computational archive. The package README is the authoritative list of commands and file roles.

[write] **5 October 2026.** The package README, SHA256SUMS and the `foundation/` copies are not shipped; `SOURCE_PROVENANCE.json` is shipped as `data/148-growing-SOURCE_PROVENANCE.json` and names seven source notes that were never part of the delivery. The companion is shipped with the prefix `148-growing-`. The recurrence (24.1) is Part I’s (11.1). At the write (Python 3.14.4, Windows, fresh extraction) 24 of the 26 tests passed; the other two exercise the POSIX-only output hardening and stop with “safe certificate creation requires POSIX `O_NOFOLLOW/O_DIRECTORY`”, as the package README says they require; `--verify` passed in normal and optimized mode.

The companion separates exact finite checks from analytic theorems. It compares fixed-exponent Euler rows computed from logarithmic recurrence with an independent product convolution. For integer t ,

$$nA_\varepsilon(n, t) = \sum_{d=1}^n B_\varepsilon(d, t)A_\varepsilon(n - d, t), \quad B_\varepsilon(d, t) = \sum_{j|d} \varepsilon^{d/j-1} j^{t+1}. \quad (24.1)$$

Both signs and all three output residues are tested; the exponent stays fixed across the row. Semantic fixtures check the distinction between unrestricted and distinct colored parts, the residue map in (16.4), and the shifted map in (20.2). Independent rational polynomial identities check the pure coefficients and the inverse coefficient equations. Rational comparisons verify the strict exponent inequalities without rounded logarithms. Finite falling-factorial tests cover the cutoff convention and elementary bounds.

Tests use explicit checks that remain active under Python’s optimized mode. They use exact integers and rational arithmetic rather than floating-point tolerances for the identities claimed exact. The certificate and its semantic fixtures are independently recomputed by the documented verification command. The build script uses a clean local TeX environment, disables shell escape, clears the default font map before adding the required maps, repeats until cross-references stabilize, and rejects warnings or box defects in the final settled log. The fixed allowlist archive uses normalized metadata and includes SHA-256 hashes for all payloads.

What these checks establish. They detect finite algebra, normalization, recurrence, inverse-coefficient, and packaging errors. They do not prove the uniform remainder, control an infinite tail by sampling it, establish exact-ratio monotonicity, or certify a finite- m level bracket. Those conclusions, where asserted, come from the proofs in this report. The distinction remains important even when all tests pass.

25 Background and limits of the comparison

Report 147 [7] develops the diagonal asymptotics and the fixed-compact defect crossover. Its normalized pure functions provide the natural reference scale here. The present report proves its own normalization and estimates, so the moderate-window argument can be read without importing a fixed-compact error constant.

Independent-process conditioning is classical. Arratia and Tavaré [8] describe independent component variables conditioned on a weighted sum, across assemblies, multisets, and selections.

The arXiv posting is from 2013 and identifies itself as nearly identical to the 1994 article in *Advances in Mathematics*, volume 104, pages 90–154. Arratia and DeSalvo [9] study Poisson approximations when both total size and component count are specified, including small-rank transition behavior for assemblies. These are relevant background methods; their abstract statements are not a proof of the specialized j^t Euler theorem above.

Only the abstracts and bibliographic metadata of these two primary sources were inspected for this report. No full-text theorem-by-theorem overlap review or comprehensive current priority search was performed. Thus the contribution claimed here is the explicit proved extension relative to Report 147, not a global assertion of novelty. No external submission, repository change, or contact is part of this report.

[write] 5 October 2026. The write did not read the two cited papers either. The conditioning device of Arratia and Tavaré is also used, through its Theorem 3, in the repository report `a033552-catalan-partitions`.

26 Further questions

The proof suggests several well-defined extensions, none of which is claimed as established here.

1. **Beyond bounded moderate activity.** Determine a uniform range with $s \rightarrow \infty$. The present exponential Taylor and moment bounds depend on a fixed upper bound for s ; simply allowing that constant to grow does not prove a new regime.
2. **The next mixed correction.** Resolve the terms at and beyond order $m^{-3/4}$, including shifted size-five depletion and mixed pure/nonpure terms. An all-orders organization must control the infinite atom sum at every required scale.
3. **Exact local monotonicity.** Obtain derivative estimates strong enough to decide whether the exact real- t ratio has a unique crossing within the moderate window. Uniform function-value approximation alone is insufficient.
4. **Integer sampling.** Formulate an appropriate discrete level-crossing statement for integer t , retaining the fixed multiplicative spacing of s . It should specify bracketing rather than treating a rounded continuous inverse as an exact threshold.
5. **Effective finite bounds.** Make all moment and Taylor constants explicit to turn the eventual asymptotic brackets into verifiable finite-input certificates for specified S and Q .
6. **Literature overlap.** Carry out a full-text comparison of conditioned component models and variable-weight Euler products before making any independent priority or publication claim.

[write] 5 October 2026. All six questions are carried into Section 27, with Part I's.

27 [write] Further questions and research (both Parts)

This section is added in the write (5 October 2026), under Vladimir's standing rule of 4 October 2026: no unproved claim and no stated limitation of either manuscript is dropped. Each item names its source, the argument sketch the source gives, and what is missing. Part I's Section 13 and

Part II's Section 26 are printed as delivered; the items below collect them, with the non-claims that the two manuscripts state in their text. Neither manuscript states a claim without proof that this section had to move, and no claim of either manuscript was found wrong; the one refuted claim is the OEIS conjecture (Corollary 2.3), and the one corrected decimal is recorded at (15.2).

1. **Effective onsets and certified integer thresholds.** *Sources:* Part I, question 1, the “Operational boundary” after Theorem 9.1, and Theorem 4.2 (“This onset is sufficient, not sharp”); Part II, question 5 and the remark after (19.11). *Sketch:* Part I’s tail bound already has the explicit onset $n \geq 192$, and Part II’s bad-atom bound has an explicit error (19.11) and onset. *Missing:* verified onsets for the forward estimates (9.1) and (21.2), for the logarithmic-error and derivative bounds used in Theorems 9.1 and 22.1, explicit moment and Taylor constants in Part II, and an interval-validated root solver; together they would turn the asymptotic brackets into finite-input certificates. A sharp onset in Theorem 4.2 is also open.
2. **All orders in the two crossover regimes.** *Sources:* Part I, question 2; Part II, question 2 and its Scope (“No ... all-orders theorem”). *Sketch:* Part I’s Theorem 10.1 has two algebraic corrections and the size-five term, from the Bonferroni expansions (10.22)–(10.23); Part II’s Theorem 15.1 stops at $O_S(m^{-3/4})$. *Missing:* the falling-factorial corrections to every order together with all nonpure atom families, equal decay phases combined, and in Part II’s range the shifted size-five depletion and the mixed pure/nonpure terms at and beyond $m^{-3/4}$, with control of the infinite atom sum at every scale.
3. **Beyond $z \leq Sm^{3/4}$, and the bridge to fixed exponents.** *Sources:* Part I, question 3, re-scoped (dated note there); Part II’s Theorem 15.1 answers it for $0 < z \leq Sm^{3/4}$, uniformly; Part II, question 1. *Sketch:* Part II’s proof uses exponential Taylor and moment bounds that depend on the fixed bound S ; Part I notes that substitution into either existing regime “is not justified”. *Missing:* a uniform range with $s = z/m^{3/4} \rightarrow \infty$, where the attenuation $e^{-8s^{4/3}/9}$ is no longer bounded below, and estimates connecting it with the fixed-exponent weighted-partition asymptotics that Part I attributes to Kotěšovec [5].
4. **Exact monotonicity and uniqueness of crossings.** *Source:* Part II, question 3 and the paragraph after Theorem 22.1 (“No monotonicity or uniqueness of the exact ratio has been proved”; no first crossing over all $t > 0$; no exclusion of crossings outside J_m). *Sketch:* the smooth model is strictly decreasing on the window. *Missing:* derivative estimates for the exact real- t ratio, which uniform approximation of values does not give.
5. **Integer exponents.** *Source:* Part II, question 4, and the remarks after Corollary 15.2 and Theorem 22.1; Part I’s remark after Theorem 10.1 (no fixed rounding rule for τ is claimed to realize every limit). *Sketch:* a unit change of t multiplies s by $8/9$. *Missing:* a discrete level-crossing statement for integer t that brackets rather than rounds the continuous inverse.
6. **Other integer weights.** *Source:* Part I, question 4. *Missing:* the phase spectra for other weight sequences, in particular when two part sizes compete for the integer maximum or the maximizing size changes.
7. **Convergence and canonical interpolation.** *Source:* Part I, after Theorem 7.2 (“convergence of this infinite power-log series is not claimed”) and after Proposition 8.2 (no “canonical convergent infinite smooth interpolation of the discrete sequences”). *Missing:* whether the power-log series (7.5) converges or is resummable, and whether a canonical smooth interpolation exists whose inverse carries the exponentially small layers.

8. **Literature and priority.** *Sources:* Part I, Section 12 (bounded search, “not proof that no earlier solution exists”); Part II, question 6 and Section 25 (abstracts of Arratia–Tavaré and Arratia–DeSalvo only). *Missing:* a full-text comparison with conditioned component models and variable-weight Euler products, and a search for an earlier statement of the limit $3^{1/3}$. The write did not read [5], [8] or [9] either.

References

- [1] OEIS Foundation, *A252782*. <https://oeis.org/A252782>. Definition and conjectured root limit; inspected 3 October 2026, retrieved crawl six days earlier.
- [2] OEIS Foundation, *A270917*. <https://oeis.org/A270917>. Definition and conjectured root limit; inspected 3 October 2026, retrieved crawl six days earlier.
- [3] OEIS Foundation, *A144048*. <https://oeis.org/A144048>. Parent Euler-transform array and fixed-column reference; inspected 3 October 2026.
- [4] OEIS Foundation, *A284992*. <https://oeis.org/A284992>. Distinct-color parent array; source review on 3 October 2026.
- [5] Václav Kotěšovec, *A method of finding the asymptotics of q -series based on the convolution of generating functions*, arXiv:1509.08708, 2015; version 3, 7 March 2016, printed pp. 21–22. <https://arxiv.org/abs/1509.08708>.
- [6] NIST Digital Library of Mathematical Functions, §5.11, especially the digamma and shifted log-Gamma asymptotic expansions (5.11.2) and (5.11.8), and its remainder discussion. <https://dlmf.nist.gov/5.11>. Inspected 3 October 2026.
- [7] *Diagonal Euler transforms: Exact phases, critical crossover, and residue aware inverses*. Report 147, 3 October 2026. Unchanged PDF and TeX reference copies are included in the source archive under `foundation/`. `[[write]]` Printed in this report as Part I; the included copies are byte-identical to it and are not shipped.]
- [8] R. Arratia and S. Tavaré, *Independent Process Approximations for Random Combinatorial Structures*, *Advances in Mathematics* 104 (1994), no. 1, 90–154. Author-posted version, 15 August 2013. [arXiv:1308.3279](https://arxiv.org/abs/1308.3279). Abstract and bibliographic metadata inspected 3 October 2026.
- [9] R. Arratia and S. DeSalvo, *Poisson and independent process approximation for random combinatorial structures with a given number of components, and near-universal behavior for low rank assemblies*, 2016, version 2, 4 July 2016. [arXiv:1606.04642v2](https://arxiv.org/abs/1606.04642v2). Abstract and bibliographic metadata inspected 3 October 2026.
- [10] `[[write]]` *Transseries and inversion*, ProveIt volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: Definition `p0:def:model`, Lemma `p0:lem:alpha-reduction`, Definition `p0:def:core`, Theorems `p0:thm:perturbed-inversion`, `p0:thm:core-reversion`, `p0:thm:lambert-core`, `p0:thm:staircase` and Proposition `p0:prop:factorial-core`.